Explaining the Unexplained: Evaluating XAI Faithfulness and Stability in a Continuously-Updating Airline Propensity Model

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Summary

Title: Explaining the Unexplained: Evaluating XAI Faithfulness and Stability in a Continuously-Updating Airline Propensity Model.

Abstract: Companies increasingly rely on machine-learning models to personalise offerings to individual customers, scoring each customer in real time to choose the Next-Best-Action. Automated prioritisation of these offers optimises business value such as click-through and conversion rates, but the models that estimate the underlying probabilities are often perceived as black boxes by marketing and governance stakeholders. This research investigates how offer-level propensity models in a large European passenger airline can be explained in a way that is faithful, stable, and useful to those stakeholders. Treating the production model as a black-box prediction function, the study uses three post-hoc methods, SHAP-based and LIME-based feature attributions and a shallow interpretable decision tree, to produce feature importances at the global level. These explanations are assessed for faithfulness and stability across splits that matter in this industry: over time, to capture seasonality, and across subgroups defined by flight-level data (route) and customer-level data (customer culture). The aim is to see whether the model relies on different features in different splits. The outcome is a structured way to evaluate these explanation methods that treats the airline environment as a stress test for standard XAI assumptions (feature correlation, exposure bias, and a continuously updating model), and provides evidence-based guidance on which methods are more and less reliable under those conditions, rather than a claim that any single method can be trusted outright. It also shows that, where the explanations can be trusted, they yield concrete insights for the marketing, strategy, and governance stakeholders who operate in a real production environment.

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Chapter 1

Introduction

1.1 Motivation and Background

Large organisations increasingly automate how they personalise customer interactions at scale. In industries such as telecommunications, retail banking, and aviation, companies rely on real-time decisioning systems that choose the Next-Best-Action (NBA) for each customer: which offer, message, or service to present, to whom, and when. Central to these systems are propensity models, predictive models that estimate how likely a customer is to respond positively to a given offer. The resulting score is combined with contextual weighting and economic-value levers to drive fully automated offer prioritisation(PEGA, 2025). A prominent commercial example is Pega, a low-code enterprise software platform widely used by large organisations to automate workflows and customer engagement; its Customer Decision Hub (CDH) is the real-time, always-on decisioning component that scores customers across channels (web, mobile, call centres, and email) and selects these Next-Best-Actions. While such systems have been shown to improve commercial outcomes, their predictive components are frequently complex, non-linear models that offer little transparency into why a given score was produced. This creates concrete problems for marketing and governance stakeholders, who must trust, audit, and act on model outputs without being able to inspect the reasoning behind them. It is precisely this tension between predictive power and interpretability that has driven the rapid growth of Explainable AI (XAI) as a field (Vilone & Longo, 2020), and that motivates the present research.

Explainable AI (XAI) has been extensively studied in the context of predictive modelling, with methods such as SHapley Additive exPlanations (SHAP) (S. Lundberg & Lee, 2017) and Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016) widely used to attribute feature contributions in black-box models. Existing research primarily evaluates these methods in static supervised learning settings, often assuming independently and identically distributed (i.i.d.) data and limited feature dependence(Molnar, 2025; Vilone & Longo, 2020). Under these conditions, explanations are typically assessed in isolation from the data-generating process, using proxy metrics such as fidelity and sparsity. These assumptions are frequently violated in production decisioning systems, where customers are scored in real time and the data available to evaluate explanations is a by-product of the system's own past decisions rather than a clean experimental sample. In such a setting the propensity model operates on observational data shaped by prior system decisions, on feature spaces with substantial correlation, and on a model architecture that updates continuously as new evidence accumulates. Even when analysis is restricted to a single offer-level model, these characteristics induce distributional shifts and dependency structures that may degrade the reliability of post-hoc explanations. Whether established XAI methods remain faithful and stable under these conditions is the central empirical question this thesis investigates.

The empirical context for this research is a large European passenger airline that uses Pega CDH to deliver personalised marketing through its digital channels. The system generates Next-Best-Actions by scoring individual customers against a portfolio of offers, with each group of similar offers governed by its own propensity model. The scoring is performed by Pega's Adaptive Decision Manager (ADM), the platform's self-learning component: ADM fits a Naive Bayes model for each offer and updates it continuously through online learning as new customer responses arrive, so the model evolves over time rather than being trained once and frozen. Historical decisioning logs constitute the primary dataset, containing, for each customer interaction, the candidate offers and the selected action, the propensity scores per offer, customer and loyalty features, travel and booking-context signals such as route and time-to-departure, and observed outcomes such as clicks and booking conversions. This setting reproduces precisely the conditions that motivate this research: features are domain-specific and correlated, outcomes are exposure-biased by prior system decisions, distributions shift across seasons and route categories, and a separate propensity model operates per offer. The primary empirical focus is a single offer model within this portfolio, a 15 kg luggage add-on scored in the Email/Outbound channel, referred to throughout as L5B15. To test whether the findings generalise, three further offer models are included as replication cases: one other luggage offer that shares a similar feature space, and two third-party partner offers (accommodation and car rental) whose feature spaces and commercial contexts differ more substantially. Each is governed by its own independently-updating ADM instance. A direct consequence of ADM's continuous updating is that the model is a moving target: during the observation window, more than 1,500 distinct model versions were active across the L5B15 scoring events alone, meaning different customers were scored by different model snapshots. Rather than treating this as a data-quality problem to be filtered away, this thesis treats continuous model updating as a defining and realistic property of production decisioning systems, one that the XAI literature has not previously examined empirically. The airline context therefore serves not merely as a business application but as a suitable testing environment for evaluating XAI methods under conditions that standard benchmark datasets systematically fail to replicate.

Evaluating explanations in production CDH environments is non-trivial for three reasons. First, real customer and behavioural data contain substantial feature correlations. In an airline setting, variables such as route, destination, and trip duration may all encode the same underlying travel pattern, causing attribution methods to assign importance arbitrarily across correlated features rather than isolating the true drivers (S. Lundberg & Lee, 2017). This makes faithfulness difficult to assess and easy to inflate. Second, CDH decisioning data is inherently observational and exposure biased. Customers can only respond to the offers they are actually shown, meaning observed conversion outcomes reflect historical system behaviour rather than true customer preferences. Both propensity models and their explanations are therefore at risk of encoding system induced patterns rather than genuine behavioural signals. This creates a feedback loop that complicates any causal interpretation of feature attributions (Lundberg, 2018). Third, the multi-model architecture of CDH systems means that explanation instability may arise from three distinct sources: sensitivity of the explanation method itself, shifts in the feature distribution across time or segments, and changes in the underlying model due to continuous online updating. This thesis proposes a structured diagnostic that distinguishes these sources by tracking, for each comparison, how much the propensity-score distribution shifts and how much the underlying model changes, and relating both to the observed change in explanation rankings. This three-way decomposition has, to our knowledge, not previously been operationalised for a production decisioning platform.

This thesis addresses these challenges by providing a structured empirical evaluation of post-hoc explanation methods applied to offer-level propensity models within a real-world airline CDH environment. Treating the production system as a black-box prediction function, we compare three explanation families representing fundamentally different approaches to explanation: additive attribution, local approximation, and global approximation. Specifically, we study SHAP, which decomposes predictions into feature contributions; LIME, which approximates model behavior locally around individual predictions; and rule-based surrogate models, which summarize overall model behavior through interpretable decision rules. The production propensity model is accessed exclusively through its logged output scores. In practice the live model cannot be queried directly and its internal parameters are not exposed; only the scores it recorded for past decisions are available. This reflects the realistic constraint practitioners face when a vendor platform logs what each model predicted but does not expose the model itself for inspection or re-scoring. Crucially, the model is not treated as fixed: the continuous updating behaviour of Pega ADM is incorporated into the evaluation design rather than controlled away.

Because the observational and exposure-biased nature of CDH data makes causal ground truth inaccessible, evaluation focuses on fidelity, defined as how accurately explanations reflect the model's predictions, and stability across temporal splits, route subgroups, and customer culture cohorts. Finally, observed explanation instability is not automatically blamed on the explanation method: by separately tracking how much the propensity-score distribution shifts and how much the underlying model changes between splits, the observed instability is decomposed into three sources, explainer sensitivity, feature-distribution shift, and model-version churn.

1.2 Research Problem and Questions

This thesis investigates whether post-hoc explanation methods produce faithful and stable outputs when applied to a continuously self-updating propensity model in a production airline CDH environment, and proposes a diagnostic protocol for attributing observed explanation instability to its source.

This problem is addressed through five research questions:

RQ1 How accurately do the prediction models underlying each explanation method reproduce the Pega propensity scores on a held-out test set, as measured by a suite of rank-based, pointwise, and distributional fidelity metrics?

RQ2 How consistent are the feature rankings produced by each explanation method across temporal, route, and customer culture splits, as measured by Spearman rank correlation and Jaccard similarity?

RQ3 When feature attribution rankings shift across splits, what is the dominant source: changes in the underlying Pega model, shifts in the input data distribution, or sensitivity of the explanation method itself?

RQ4 Do the fidelity and stability patterns observed for the primary luggage offer model replicate across three further offer variants, including a generic luggage offer and two third-party partner offers in different product categories, each governed by its own independently-updating Pega instance?

RQ5 Which features most strongly drive the offer-level propensity scores, how does importance distribute across feature groups, and what do the resulting attributions imply for marketing strategy and data governance in a production CDH?

Where RQ1–RQ4 evaluate the explanation methods themselves, RQ5 turns to the *content* of the explanations: what the production model has in fact learned to score on, and what that implies for the marketing and governance stakeholders who must act on its outputs.

1.3 Contributions

The starting point is a practical problem. A practitioner who wants to explain a production propensity model can choose from many explanation methods, and these methods do not always agree with one another, which makes it hard to know which explanation to trust. This thesis takes three of the most widely used methods, SHAP, LIME, and an interpretable decision-tree surrogate, and asks not only whether their explanations are faithful and stable, but also, when their explanations change across populations, *why* they change: because the data shifted, because the underlying model changed, or because the method itself is sensitive.

From this, the thesis makes contributions at two levels. Methodologically, we propose a reusable protocol for evaluating explanation fidelity and stability in observational decisioning environments, combining surrogate approximation error with a three-way instability attribution. The core methodological contribution is this attribution step: it decomposes an observed shift in explanation rankings into three sources, explainer sensitivity, a shift in the feature distribution, and churn in the underlying model, by tracking how much the propensity-score distribution and the set of active model versions change between splits. This lets a practitioner assign instability to its source rather than conflating the three, which to our knowledge has not previously been operationalised in this form for a production decisioning platform. The findings are then replicated across three further offer models, each governed by its own independently-updating ADM instance, giving a generalisation check on whether explanation fidelity and stability are specific to one offer or hold more broadly: both for an offer that shares a similar feature space and for offers whose feature spaces differ more substantially.

Empirically, the study provides the first systematic comparison of post-hoc explanation methods applied to Pega ADM propensity scores under production conditions, reporting fidelity and stability under the specific combination of continuous model updating, categorical-heavy feature spaces, and sparse interaction histories that characterise real deployments. Beyond the methodological evaluation, the study reports the substantive attribution findings themselves: which feature groups drive the offer-propensity scores, and two implications of direct commercial relevance, namely that the model scores recency and contact-fatigue rather than customer identity, and that one of its most influential predictors encodes data missingness rather than behaviour.

1.4 Thesis Outline

The remainder of this thesis is organised as follows. Chapter 2 presents the data and methodology. It describes the airline decisioning logs and the offer portfolio, formalises the black-box surrogate setup, and defines the surrogate model, the three explanation methods, the fidelity and stability metrics, the experimental splits, and the instability-attribution procedure. Chapter 3 reports the results, organised by research question: surrogate fidelity (RQ1), explanation stability (RQ2), instability attribution (RQ3), cross-offer replication (RQ4), and the feature-level drivers and their business implications (RQ5). Chapter 4 interprets these findings in relation to the XAI literature, draws out the academic and practical implications, and discusses the assumptions and limitations of the study. Chapter 5 concludes and outlines directions for future research.

Chapter 2

Methodology

Figure 1 provides a schematic overview of the full methodology before each component is formalised in the sections that follow.

2.1 Problem Formulation

Let $\mathbf{X} \in \mathcal{X} \subseteq \mathbb{R}^d$ denote the vector of customer and context features, and let the Pega Adaptive Decision Model define a black-box prediction function

$$f : \mathcal{X} \to [0, 1], \quad p = f(\mathbf{X}),$$

where p is the *propensity score*: Pega's term for the model's estimated probability that the customer responds positively to (clicks on) the offered action at the moment of scoring. The model is implemented as a Naive Bayes classifier updated via online learning, but its internal parameters and likelihood components are not accessible. Instead, only logged tuples $(\mathbf{X}_i, p_i)$ are observed from historical decision data, where the index $i = 1, \ldots, n$ runs over the individual logged scoring events (each one customer scored against the offer at a single point in time) and n is the total number of such events. As a consequence, the model is treated as a black-box function evaluated offline, without access to a live scoring API.

Because direct queries to f are unavailable, all explanation methods operate through a surrogate model g trained to approximate f on the observed data:

$$g \approx f \quad \text{based on } \{(\mathbf{X}_i, p_i)\}_{i=1}^{n}.$$

Feature attributions are then obtained as

$$\boldsymbol{\phi}(\mathbf{X}) = (\phi_1(\mathbf{X}), \ldots, \phi_d(\mathbf{X})) = \text{Explain}(g, \mathbf{X}),$$

inducing the approximation pipeline $f \to g \to \boldsymbol{\phi}$, where explanation quality depends jointly on surrogate accuracy and explainer behaviour. Explanations therefore target g rather than f directly, and their validity as descriptions of the black-box model is dependent on how accurately g approximates f.

This assumption does not hold in general: models with similar global predictive accuracy may rely on different feature interactions, producing divergent explanations. The surrogate is therefore evaluated not only on prediction error but on rank preservation and distributional alignment of p, as these jointly determine whether g captures the aspects of f that matter for explanation. Results throughout should be interpreted as explanations of g, with validity as a proxy for f depending on these approximation diagnostics.

Fidelity is defined as the agreement between the black-box model f and the surrogate model g on held-out observations. Let

$$\mathbf{p} = (p_1, \ldots, p_n) \quad \text{and} \quad \mathbf{q} = (q_1, \ldots, q_n),$$

where $p_i = f(\mathbf{X}_i)$ and $q_i = g(\mathbf{X}_i)$, i.e. the logged and surrogate-predicted propensity scores. Surrogate fidelity is then defined abstractly as

$$\text{Fidelity}(f, g) = A(\mathbf{p}, \mathbf{q}),$$

where A is an agreement function. Since agreement may concern pointwise accuracy, rank preservation, or distributional alignment, A is instantiated through multiple complementary metrics rather than a single loss function. Formal definitions are provided in Section 2.5.

The second evaluative dimension is stability: whether explanation outputs are consistent across the conditions under which the system operates. Let π_t denote the feature importance ranking over split t, where features are ordered by their mean absolute importance score, with the specific importance measure defined by each explanation method. Stability is then defined as the consistency of π_t across splits:

$$\text{Stability}(\pi_{t_1}, \pi_{t_2}) = S(\pi_{t_1}, \pi_{t_2}),$$

where S is instantiated as Spearman rank correlation ρ_S for full-ranking consistency and Jaccard similarity at cutoff k for top-k overlap. Specific definitions are provided in Section 2.5.

Observed instability $\Delta\pi = 1 - \rho_S(\pi_{t_1}, \pi_{t_2})$ may arise from three distinct sources: shifts in the input distribution $\mathcal{D}$ across splits (distribution shift, δ_d), changes in f due to continuous online updating of the Pega ADM model (model churn, δ_m), or sensitivity of the explanation method itself to distributional variation (explainer sensitivity, δ_e), such that observed instability $\Delta\pi$ may reflect the joint influence of δ_d, δ_m, and δ_e. Section 2.5 operationalises a diagnostic that attributes $\Delta\pi$ across these three sources, constituting the primary methodological contribution of this thesis.

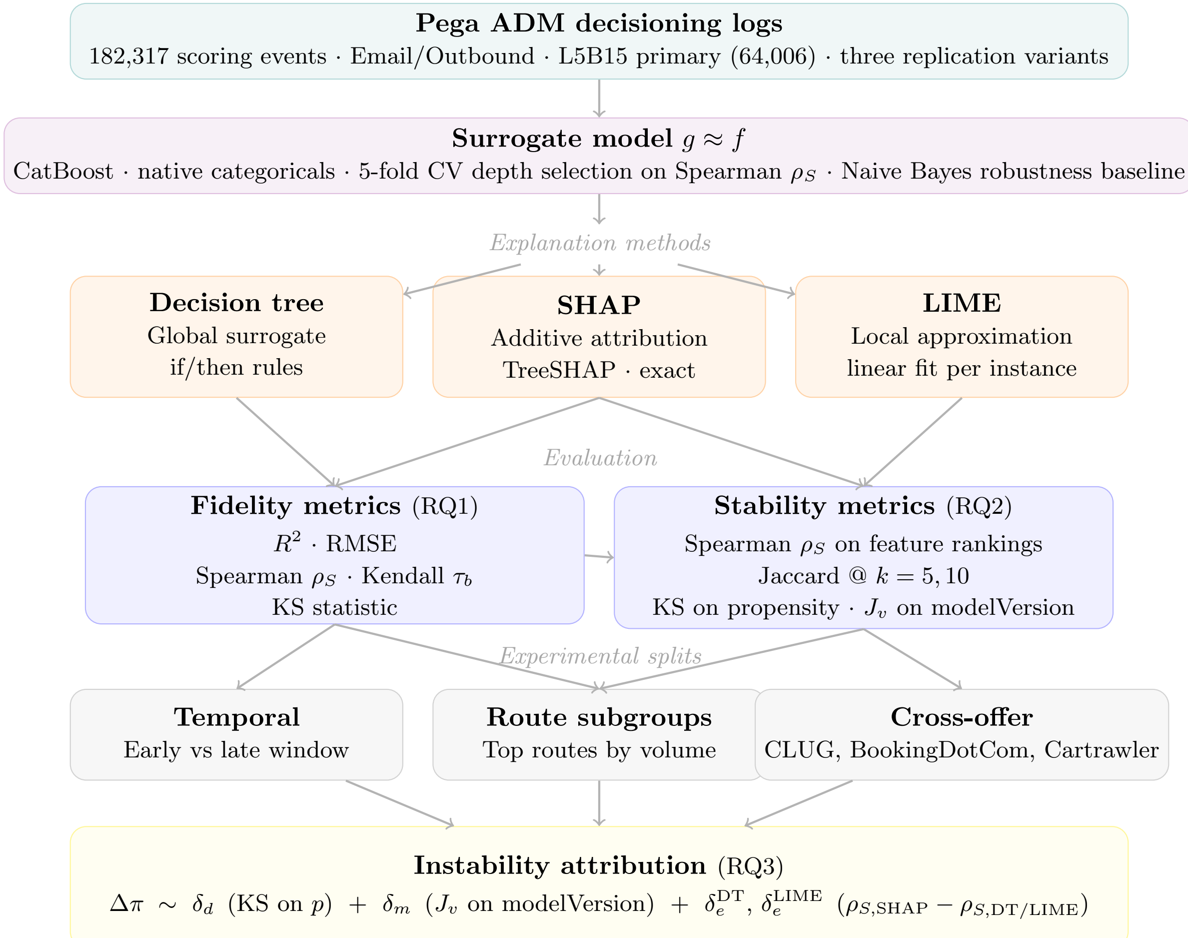

Figure 1: Overview of the methodology. Pega ADM decisioning logs are used to train a surrogate model g, which serves as the prediction oracle for three explanation methods. Each method is evaluated on fidelity against logged propensity scores and on stability across experimental splits, including a cross-offer replication over four offer variants (RQ4). Observed instability is decomposed into three sources, the primary methodological contribution (RQ3). The validated attributions then feed the feature-driver analysis of RQ5 (not shown).

2.2 Data

The raw data consists of Pega CDH interaction records exported as nested JSON files from a large European passenger airline. Each interaction record contains the full decisioning context for a single customer contact, including all candidate offers evaluated, the selected action, and the propensity scores assigned to each offer. Records are parsed and deduplicated on a unique execution identifier to prevent double-counting across overlapping exports. After filtering to the Email and Outbound channel and retaining only sales actions, the working dataset contains 180,957 scoring events across four offer variants. Minor luggage variants with fewer than 2,000 observations combined are present in the raw data but excluded from analysis due to insufficient volume for stable surrogate training and split-based evaluation.

The 180,957 scoring events span four analysed variants summarised in Table 1. The primary analysis focuses on the 15 kg luggage add-on variant (L5B15, n = 64,006), selected as the highest-volume weight-specific offer and the most commercially established product in the portfolio. Three further variants are retained as replication cases: the generic luggage offer (CLUG, n = 34,230), which shares the same product context as L5B15; and two third-party partner offers, BookingDotCom (n = 40,561, accommodation) and Cartrawler (n = 42,160, car rental), which operate in entirely different product categories. Including partner offers alongside luggage variants tests whether explanation behaviour generalises not only across models with similar feature spaces but across models with different commercial contexts, providing a stronger generalisability check for RQ4 than weight-variant replication alone. Each variant is governed by its own independently-updating Pega ADM instance with a partially distinct set of active predictors, so replication is conducted within each variant's own feature space.

Table 1: Volume by offer variant. Replication variants span both luggage and third-party partner offer types.

Variant	Type	n
L5B15	15 kg luggage add-on (primary)	64,006
CLUG	Generic luggage offer	34,230
BookingDotCom	Third-party accommodation offer	40,561
Cartrawler	Third-party car rental offer	42,160

The feature space is restricted to the predictors confirmed active in the live Pega ADM model for each variant, as identified from ADM model snapshots. This restriction is deliberate: Pega ADM automatically selects and deactivates predictors according to how much predictive signal each one carries, disabling those whose association with the target falls below an internal threshold. Including inactive predictors would mean training a surrogate on features the black-box model does not use, causing explanations to reflect the surrogate's own behaviour rather than that of the model being explained. The number of active predictors varies across variants, ranging from 22 in the CLUG variant to 33 in BookingDotCom, reflecting each model's independent predictor selection. Predictors span four feature groups: booking context, flight data, contact history, and scoring parameters, with customer attributes appearing additionally in the BookingDotCom variant only. Missingness varies substantially across groups: flight data features are largely complete across all variants (mean 1.1–2.1%), while contact history features are considerably sparser (mean 41.6–57.4%), reflecting customers with limited or no prior contact history on a given channel. Scoring parameters are set by the system at scoring time rather than read from a customer record, so the notion of a missing value does not apply to them: there is no row in which the value is absent, and the missingness column therefore reports a dash rather than a percentage in the appendix table. A full breakdown of active predictor counts and missingness by feature group and variant is provided in Appendix A.1.

The target variable is the Pega ADM propensity score $p \in [0, 1]$, representing the model's estimated probability that the customer responds positively to (clicks on) the offer at the moment of scoring. Observed outcomes are not used as the target: the goal of this study is to explain what drives Pega's model output, not to model customer behaviour directly. Using observed outcomes would require assumptions about exposure and selection that cannot be satisfied with observational decisioning data. A further structural characteristic of the dataset is that the underlying model is not static: Pega ADM updates its Naive Bayes model continuously via online learning, meaning different customers in the observation window were scored by different model snapshots. Each scoring event is logged with a model-version identifier denoting which snapshot produced the propensity score, alongside descriptive metadata on the number of customer responses and the model's AUC at the time. The model-version field is excluded from the surrogate feature matrix but retained as the diagnostic proxy for model version churn in the instability attribution analysis in Section 2.5.

2.3 Surrogate Model

Because direct queries to the Pega ADM model f are unavailable, all explanation methods operate through a surrogate model g trained to approximate f on the observed logged scores. The surrogate serves two roles: as the prediction oracle required by SHAP and LIME, and as the first stage of fidelity evaluation, since the quality of any downstream explanation is bounded by how accurately g replicates f's input-output mapping.

Rather than fixing the surrogate architecture on theoretical grounds alone, five candidate models are compared empirically on the L5B15 dataset using the full suite of fidelity metrics defined in Section 2.5, with the final selection based on performance across all five metrics jointly. The candidates represent a range of model complexity and different approaches to handling categorical features: linear regression as a lower bound, Gaussian Naive Bayes as an architecture-matched baseline given that Pega ADM uses Naive Bayes internally, Random Forest and LightGBM as alternative flexible models, and CatBoost as the proposed surrogate. All five candidates are evaluated on the same stratified 80/20 train/test split, stratified on propensity decile bins to ensure the test set covers the full propensity range, which is important for rank-based and distributional metrics. The full comparison is reported at the start of Section 3.

Beyond rank-based fidelity, two additional properties guide the final architecture choice. The first is compatibility with TreeSHAP, the SHAP implementation used in this study. TreeSHAP computes exact Shapley values analytically by exploiting the tree structure of the model, which is only possible for tree-based architectures. Applying SHAP to a non-tree surrogate would require KernelSHAP instead, a model-agnostic approximation that introduces its own sampling variance on top of the surrogate approximation error, compounding exactly the sources of inexactness the study aims to isolate. The second property is categorical encoding. Several candidates require ordinal encoding as a preprocessing step, which assigns arbitrary integer values to categorical levels and introduces a numeric ordering that has no real-world meaning. A surrogate that handles categoricals natively avoids this: it learns directly from the categorical values themselves, meaning that SHAP attributions reflect the true categorical relationships driving propensity scores rather than an arbitrary encoding artefact. Missing values in sparse features such as contact history are handled in the same way, preserving absence of information as a meaningful signal consistent with how Pega ADM treats its own missing predictor bins.

Tree depth is treated as a tuning parameter selected independently per variant via five-fold cross-validation on Spearman ρ_S over depths 1 to 15, using 200 boosting iterations during the depth sweep and 500 for the final fit. The reduced iteration budget during cross-validation is sufficient because the relative ranking of depths stabilises well before convergence. Depth is the most influential complexity control for CatBoost's symmetric oblivious trees, so it is the only hyperparameter tuned; the remaining settings, including the learning rate, the number of boosting iterations, and the L2 leaf regularisation, are held at their library defaults, which keeps the surrogate-fitting protocol simple, reproducible, and identical across all four offer variants. Breiman's 1-SD rule is applied to the resulting CV curve (Hastie et al., 2009): the selected depth is the smallest value whose mean ρ_S falls within one fold-to-fold standard deviation of the CV maximum, operationalising the principle that when several depths are statistically indistinguishable from the best, the simplest should be preferred. The grid upper bound of 15 is chosen to extend past CatBoost's typical effective depth on tabular data by a sufficient margin to confirm that the mean-ρ_S curve plateaus inside the grid rather than at its edge, which would indicate a range-cap artefact rather than a genuine optimum. Selected depths and final fidelity results per variant are reported in Section 3.

2.4 Explanation Methods

Three post-hoc explanation methods are evaluated, representing fundamentally different approaches to interpreting black-box model predictions: additive feature attribution via SHAP, local linear approximation via LIME, and global rule-based approximation via a shallow decision tree surrogate. All three methods operate on logged propensity scores $p_i = f(\mathbf{X}_i)$ rather than a live scoring API, reflecting the realistic constraint that Pega ADM's internals are inaccessible. SHAP and LIME both require a callable prediction function and therefore use the CatBoost surrogate g as their oracle, inducing the two-layer approximation pipeline $f \to g \to \boldsymbol{\phi}$ established in Section 2.1. The decision tree surrogate is fitted directly on p_i and does not require an intermediate model, introducing only a single layer of approximation error.

2.4.1 Global Surrogate: Decision Tree

A shallow regression tree is fitted directly on the logged propensity scores $p_i = f(\mathbf{X}_i)$, providing a globally interpretable approximation of the black-box model's input-output mapping. Tree depth is treated as a hyperparameter and selected by five-fold cross-validation optimising Spearman rank correlation over the range one to ten, consistent with the surrogate model depth selection described in Section 2.3. Because the decision tree is implemented with scikit-learn, the standard Python machine-learning library, rather than CatBoost, categorical features are encoded using ordinal encoding prior to fitting. This encoding choice is a practical constraint of the sklearn interface rather than a modelling preference; its potential effect on feature importance rankings relative to the CatBoost-based methods is acknowledged as a limitation in Section 4.5. Decision rules are extracted as human-readable if/then conditions, producing a compact summary of the dominant decision logic governing propensity scores across the full dataset. Global feature importance is then read directly from the fitted tree as the total reduction in squared error contributed by all splits on each feature, normalised across features to sum to one:

$$\mathrm{Imp}(j) = \frac{\sum_{t \in \mathcal{T}_j} n_t \, \Delta_t}{\sum_{j'=1}^{d} \sum_{t \in \mathcal{T}_{j'}} n_t \, \Delta_t},$$

where $\mathcal{T}_j$ is the set of internal nodes that split on feature j, n_t is the number of training instances reaching node t, and Δ_t is the reduction in squared error achieved by the split at node t. These normalised importances are the decision tree's analogue of the SHAP and LIME importance scores and are what enter the stability comparison. Fidelity is assessed using the metrics defined in Section 2.5.

2.4.2 SHAP

SHAP (SHapley Additive exPlanations) decomposes each prediction into an additive sum of feature contributions grounded in cooperative game theory (S. Lundberg & Lee, 2017). For a prediction $q_i = g(\mathbf{X}_i)$, SHAP instantiates the generic attribution vector $\boldsymbol{\phi}$ from Section 2.1 by assigning each feature j a Shapley value $\phi_j(\mathbf{X}_i)$ such that

$$q_i = \phi_0 + \sum_{j=1}^{d} \phi_j(\mathbf{X}_i),$$

where $\phi_0 \in \mathbb{R}$ is the expected model output over the training data, $\phi_0 = \mathbb{E}[g(\mathbf{X})]$, and $\boldsymbol{\phi}(\mathbf{X}_i) = (\phi_1(\mathbf{X}_i), \ldots, \phi_d(\mathbf{X}_i))$ is the vector of instance-level attributions introduced in Section 2.1. The completeness axiom guarantees that attributions sum exactly to the model output, providing an internally consistent decomposition by construction. SHAP values are computed using TreeSHAP, which exploits the tree structure of the CatBoost surrogate to derive exact Shapley values analytically, avoiding the sampling variance introduced by model-agnostic approximations such as KernelSHAP (S. Lundberg & Lee, 2017). Global feature importance is summarised as mean absolute SHAP values aggregated across all instances. Because SHAP values are computed on g rather than f directly, explanation fidelity inherits from surrogate fidelity, assessed using the metrics defined in Section 2.5.

2.4.3 LIME

LIME (Local Interpretable Model-agnostic Explanations) approximates model behaviour in the neighbourhood of individual predictions using a locally fitted linear model (Ribeiro et al., 2016). For a given instance $\mathbf{X}_i$, LIME generates a set of perturbed samples in the vicinity of $\mathbf{X}_i$, queries the surrogate g to obtain predicted propensity scores for each perturbed sample, and fits a weighted linear regression to the resulting input-output pairs. The linear coefficients serve as local feature importances for that instance. Formally, the local explanation for $\mathbf{X}_i$ is obtained as

$$\xi(\mathbf{X}_i) = \arg\min_{h \in H} \mathcal{L}\big(g, h, \pi_{\mathbf{X}_i}\big) + \Omega(h),$$

where H is the class of linear models $h(\mathbf{z}) = \beta_0 + \sum_{j=1}^{d} \beta_j z_j$, $\pi_{\mathbf{X}_i}$ is a proximity kernel that weights each perturbed sample by its closeness to $\mathbf{X}_i$, $\mathcal{L}$ is the weighted squared-error loss of h against the surrogate g on those samples, and Ω penalises explanation complexity. The fitted coefficients β_j are the local feature importances $\phi_j(\mathbf{X}_i)$ for that instance. In this study, the CatBoost surrogate serves as the prediction oracle: LIME perturbs instances and queries g rather than Pega's live API, which is consistent with the black-box access constraint and ensures that all explanation methods operate under identical conditions. LIME explanations are computed for a sample of 500 instances with 200 perturbations per instance, balancing computational cost against coverage of the feature space. LIME's perturbation mechanism produces binned condition strings rather than raw feature names; these are mapped back to base feature names before aggregation. Global importance is summarised as mean absolute LIME coefficients across all sampled instances. For SHAP and the decision tree surrogate, importance scores are well-defined globally and aggregation is straightforward. For LIME, this aggregation assumes coefficient comparability across instances, which may not hold in all regions of the feature space and is treated as a limitation in Section 4.5.

2.5 Evaluation Framework

Explanation quality is assessed along two dimensions established in Section 2.1: fidelity, measuring how accurately each explanation method approximates the black-box propensity scores, and stability, measuring how consistently feature attribution rankings are reproduced across different evaluation splits. This section provides formal definitions of all metrics used to operationalise these dimensions.

2.5.1 Fidelity Metrics

Fidelity is assessed using four complementary metrics, each capturing a different aspect of surrogate agreement with the logged propensity scores $p_i = f(\mathbf{X}_i)$. Let $q_i = g(\mathbf{X}_i)$ denote the surrogate prediction for instance i on a held-out test set of size n.

The coefficient of determination is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(p_i - q_i)^2}{\sum_{i=1}^{n}(p_i - \bar{p})^2},$$

where $\bar{p} = \frac{1}{n}\sum_{i=1}^{n} p_i$ is the mean logged propensity score. R^2 measures the proportion of variance in p explained by g, with values closer to one indicating higher fidelity.

Root mean squared error is defined as

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(p_i - q_i)^2},$$

providing a pointwise measure of prediction error on the scale of the propensity scores themselves.

Spearman rank correlation ρ_S and Kendall's τ_b are computed between the vectors $(p_1, \ldots, p_n)$ and $(q_1, \ldots, q_n)$. Both measure rank preservation: whether the surrogate orders instances in the same way as the black-box model. Spearman's ρ_S is defined as the Pearson correlation of the rank-transformed scores:

$$\rho_S = \frac{\sum_{i=1}^{n}(R_i - \bar{R})(T_i - \bar{T})}{\sqrt{\sum_{i=1}^{n}(R_i - \bar{R})^2 \sum_{i=1}^{n}(T_i - \bar{T})^2}},$$

where R_i and T_i denote the ranks of p_i and q_i respectively, and $\bar{R}$, $\bar{T}$ are the corresponding mean ranks (the average of the rank values, equal to $(n+1)/2$ in the absence of ties). Kendall's τ_b is reported alongside Spearman's ρ_S because it explicitly corrects for tied ranks, which are common here: Pega's isotonic calibration maps many instances onto identical propensity values, so the logged scores contain substantial ties. It is defined as

$$\tau_b = \frac{|\text{concordant pairs}| - |\text{discordant pairs}|}{\sqrt{(C - T_p)(C - T_q)}},$$

where $C = \binom{n}{2}$ is the total number of pairs, T_p is the number of pairs tied on p, and T_q is the number of pairs tied on q. Rank preservation is treated as the primary fidelity criterion, as Pega CDH uses propensity scores to rank and prioritise offers rather than as calibrated probabilities, making order-consistency more relevant than pointwise accuracy for downstream decisioning.

Distributional alignment is assessed using the two-sample Kolmogorov-Smirnov statistic:

$$D_{KS} = \sup_x |F_p(x) - F_q(x)|,$$

where F_p and F_q are the empirical cumulative distribution functions (ECDFs) of the logged and surrogate propensity scores respectively. A small D_{KS} indicates that the surrogate reproduces the overall shape of the propensity score distribution.

2.5.2 Stability Metrics

Stability is defined as the consistency of feature importance rankings across evaluation splits. Let $\pi_t = (\pi_t^{(1)}, \dots, \pi_t^{(d)})$ denote the ranking of the d active predictors induced by their mean absolute importance score over split t, where rank one denotes the most important feature and the specific importance measure is defined by each explanation method. Here t_1 and t_2 denote the two splits being compared.

Full ranking consistency is measured using Spearman rank correlation between π_{t_1} and π_{t_2}, defined as the Pearson correlation of the rank vectors:

$$\rho_S(\pi_{t_1}, \pi_{t_2}) = \frac{\sum_{j=1}^{d}\left(\pi_{t_1}^{(j)} - \bar{\pi}_{t_1}\right)\left(\pi_{t_2}^{(j)} - \bar{\pi}_{t_2}\right)}{\sqrt{\sum_{j=1}^{d}\left(\pi_{t_1}^{(j)} - \bar{\pi}_{t_1}\right)^2 \sum_{j=1}^{d}\left(\pi_{t_2}^{(j)} - \bar{\pi}_{t_2}\right)^2}},$$

where $\bar{\pi}_t = \frac{1}{d}\sum_{j=1}^{d}\pi_t^{(j)}$ is the mean rank in split t. Values close to one indicate that the full feature ranking is preserved across splits; values close to zero indicate that rankings are essentially uncorrelated.

Top-k overlap is measured using the Jaccard similarity coefficient between the sets of top-k features identified in each split:

$$J_k(\pi_{t_1}, \pi_{t_2}) = \frac{|S_{t_1}^{(k)} \cap S_{t_2}^{(k)}|}{|S_{t_1}^{(k)} \cup S_{t_2}^{(k)}|},$$

where $S_t^{(k)}$ denotes the set of the k highest-ranked features in split t. Jaccard similarity is evaluated at $k = 5$ and $k = 10$: $k = 5$ captures the features a practitioner would inspect first; $k = 10$ captures broader agreement across the top half of the active predictor set. A value of one indicates perfect overlap; zero indicates no shared features in the top-k. Both metrics are computed for each explanation method separately across temporal, route-based, and culture-based splits, and reported as a summary table enabling direct comparison of stability across SHAP, LIME, and the decision tree surrogate.

To validate that observed between-split decision tree instability reflects genuine subgroup sensitivity rather than random fitting variance, a bootstrap procedure is applied: the decision tree is re-fitted ten times on the same data using different random seeds, and the spread in feature rankings is compared against the between-split instability gap. A between-split gap that exceeds the bootstrap variance confirms that instability is attributable to subgroup composition rather than fitting noise.

2.5.3 Instability Attribution

Observed instability $\Delta\pi = 1 - \rho_S(\pi_{t_1}, \pi_{t_2})$ may reflect any combination of three sources introduced in Section 2.1: distribution shift δ_d, model version churn δ_m, and explainer sensitivity δ_e. Because all three sources operate simultaneously in observational CDH data, a precise causal decomposition is not possible. Instead, this study proposes a structured diagnostic that jointly characterises each source and uses their co-occurrence pattern to attribute instability to its likely dominant cause. All thresholds are applied to effect-size statistics directly; p-values are intentionally omitted because at sample sizes in the thousands, even trivially small effect sizes reach conventional significance levels.

Distribution shift δ_d is monitored using the two-sample Kolmogorov-Smirnov statistic on the propensity score distribution across splits. KS was chosen over the Population Stability Index (PSI), a widely used drift metric that bins the score range and sums the weighted relative-frequency differences between the two samples. PSI is poorly suited here because Pega ADM maps log-odds scores to propensities via an isotonic calibration step that produces a step function: instances in the same score bin all receive the same propensity value. The resulting distribution clusters on a small set of recurring values rather than spreading continuously across $[0, 1]$, which causes PSI to be sensitive to where bin boundaries fall and to inflate artificially when bins split these clustered values unevenly. KS operates directly on ECDFs without any binning or smoothing parameter, making it robust to this artefact. The PSI bin-sensitivity analysis is documented in Appendix A.5 as a sensitivity check. A propensity distribution shift is classified as significant when $\mathrm{KS}_{\hat{p}} \geq 0.10$, a threshold corresponding to a maximum gap of ten percentage points between the two empirical distributions: a moderate effect size chosen to flag operationally meaningful shifts while ignoring negligible ones.

Model version churn δ_m is proxied by the Jaccard overlap, applying the same similarity coefficient defined above for top-k feature sets, now to the sets of active model-snapshot identifiers in each split:

$$J_v = \frac{|V_{t_1} \cap V_{t_2}|}{|V_{t_1} \cup V_{t_2}|},$$

where V_t is the set of distinct model snapshot identifiers observed in split t. A value near zero indicates that the two splits were scored by largely non-overlapping snapshots; a value near one indicates substantial overlap. Pega ADM assigns a new snapshot identifier on every micro-update of the online Naive Bayes, so J_v measures snapshot turnover rather than parameter shift directly: two functionally identical models may carry different identifiers if an update occurred between splits. This limitation is mitigated by a structural property of the route and culture splits used in this study (defined in Section 2.6). Because those splits are drawn from the same observation window, the production scorer was drawing on the same pool of active snapshots by construction, and any reasonable δ_m proxy returns a value near zero for those splits. The RQ3 conclusion is therefore robust to the choice of churn proxy, even if J_v alone is imperfect as a general parameter-shift metric. Model version churn is classified as significant when $J_v \leq 0.10$. This threshold sits in an empirically observed gap in the bimodal distribution of J_v values: temporal splits produce values near zero while same-window splits produce values near 0.5, with no observed values in the interval $(0.10, 0.40)$. A sensitivity analysis varying the threshold across this interval confirms that split classifications are invariant, as documented in Appendix A.6.

Explainer sensitivity δ_e captures how much of the observed ranking instability stems from the explanation method itself, rather than from changes in the data or the model. SHAP serves as the reference because TreeSHAP is deterministic given a fixed surrogate: a single CatBoost surrogate is fitted once on the full training set and applied identically to both splits, so any change in SHAP rankings reflects only δ_d and δ_m with no contribution from refitting or sampling variance. The decision tree and LIME explainers are then compared against this reference, probing two distinct mechanisms of explainer sensitivity:

- DT refit sensitivity, $\delta_e^{\mathrm{DT}} \approx \rho_{S,\mathrm{SHAP}} - \rho_{S,\mathrm{DT}}$, captures variance from re-fitting the surrogate tree structure independently on each split.
- LIME sampling sensitivity, $\delta_e^{\mathrm{LIME}} \approx \rho_{S,\mathrm{SHAP}} - \rho_{S,\mathrm{LIME}}$, captures variance from LIME's local perturbation sampling. LIME shares the fixed-surrogate property with SHAP, so any gap here is sampling-side, not refit-side.

A positive value indicates that the corresponding explainer is less stable than SHAP beyond what δ_d and δ_m alone would predict. The two diagnostics are complementary of the same instability construct and are reported jointly throughout the Results.

The three diagnostic measures are combined into a classification rule summarised in Table 2. Each split is assigned to its primary instability source based on the co-occurrence of the two external-source diagnostics, $\mathrm{KS}_{\hat{p}}$ and J_v. The explainer gaps δ_e^{DT} and δ_e^{LIME} are reported alongside the classification as descriptive evidence; they are not classifier inputs, but they validate an Explainer assignment when large and qualify a Distribution shift, Model churn, or Both assignment when small.

Table 2: Attribution classification rule. Each split is assigned to its primary instability source based on the two external-source diagnostics. $\mathrm{KS}_{\hat{p}}$ threshold ≥ 0.10; J_v threshold ≤ 0.10. The explainer-sensitivity gaps δ_e^{DT} and δ_e^{LIME} are reported as descriptive evidence, not thresholded.

Primary source	$\mathrm{KS}_{\hat{p}} \geq 0.10$	$J_v \leq 0.10$	Interpretation
Explainer (δ_e)	No	No	External conditions stable; residual instability attributed to the explanation method (refitting or sampling variance)
Dist. shift (δ_d)	Yes	No	Propensity distribution shifted across splits; input data or upstream pipeline changed
Model churn (δ_m)	No	Yes	Model snapshot turnover significant; underlying model updated between splits
Both ($\delta_d + \delta_m$)	Yes	Yes	Both external conditions triggered simultaneously

This pattern-based attribution does not claim to identify causes exactly, but it gives a clear and transparent basis for interpreting explanation instability in production decisioning environments.

2.6 Experimental Design

Evaluation splits are constructed across four dimensions. Temporal splits divide each variant's observation window into an early and late period at the median decision timestamp, ensuring roughly equal sample sizes in each half. Route subgroups are constructed from the three highest-volume departure-destination pairs in L5B15. As shown in Table 3, the three pairs are structured to cover three cases, a shared origin, a shared destination, and a pair that differs in both origin and destination, providing a systematic basis for assessing whether origin or destination drives explanation instability.

Table 3: Route subgroup pairs for L5B15 stability evaluation, constructed from the four highest-volume routes.

Pair	Route A	Route B	Comparison type
1	ORY→NCE	ORY→OPO	Shared origin, different destination
2	ORY→OPO	NCE→ORY	Different origin, different destination
3	NCE→ORY	TLS→ORY	Different origin, shared destination

Culture subgroups compare the two highest-volume customer culture codes in L5B15, French (FR) and Dutch (NL), providing a customer-segment dimension alongside the route-based splits. The same temporal split is applied to the three replication variants (CLUG, BookingDotCom, and Cartrawler) to test whether the stability and attribution patterns found for L5B15 generalise to offer models with different product contexts and partially distinct feature sets.

2.7 Feature-Driver Analysis

RQ1–RQ4 evaluate the explanation methods themselves: whether SHAP, LIME, and the decision-tree surrogate reproduce the Pega scores faithfully (RQ1), whether their feature rankings are stable across temporal, route, and culture splits (RQ2), how observed shifts decompose across distribution, model-churn, and explainer sources (RQ3), and whether the picture replicates across offer variants (RQ4). Those four questions give us the confidence RQ5 requires: when the three methods agree that a feature carries high importance, under the fidelities and stabilities documented in earlier chapters, the ranking can be trusted as the production model's own signal rather than an explainer artefact. RQ5 therefore proceeds to the content-level question, namely which features the production model has in fact learned to score on. The analysis is purely associative. It makes no causal claim about which features cause add-on uptake, and only describes how much importance the production model attaches to each of them. It introduces no new model and re-uses the per-feature importances produced by SHAP, LIME, and the surrogate decision tree on L5B15.

The procedure has three steps. First, each method's importances are normalised to sum to one and ranked, yielding a cross-method view of which individual predictors carry the most weight.

Predictors are then assigned to one of four feature groups, contact history, flight data, booking context, and scoring parameter, and importance is summed within group to give each group's share of total attribution per method. The group-level view is the primary summary, motivated directly by RQ2–RQ3: even where individual ranks shift modestly across splits, group-level shares are far less sensitive, because the within-group reshuffling of importance across correlated predictors, which RQ3 partly attributes to explainer sensitivity δ_e, cancels under aggregation. Group shares therefore inherit the stability evidence of RQ2 more cleanly than individual ranks do.

Second, for each predictor in the upper portion of the cross-method ranking, we ask whether a business-logical link to luggage add-on uptake is present. Recency and frequency of prior outbound contact pass this check as a contact-fatigue proxy, trip-context fields such as seat number, destination, or booking month as a trip-type or party-size proxy, and the scoring parameter because it encodes which offer is being scored. A predictor fails the check if no such link is available on its face. This step is judgmental rather than statistical and is reported transparently; its role is not to validate the model but to surface predictors whose attributed importance demands a follow-up before reaching a stakeholder.

Third, a flagged predictor is probed in the raw data to establish whether the surrogate's importance attribution tracks the field's *value* or its mere *populated-ness*. The latter mechanism is plausible in this setting because Pega ADM's predictor selection groups predictors with pairwise correlation above an internal threshold and retains only the strongest of each group; a single surviving predictor can therefore silently encode the missingness pattern of an entire dropped cluster. The probe returns to the full pre-restriction feature matrix (441 columns, 64,006 L5B15 records) and constructs three variables. Let $Z = \mathbf{1}[\text{predictor populated}]$ denote the binary populated-indicator for the flagged predictor (chosen to avoid conflict with the feature-vector $\mathbf{X}$ used elsewhere in the methodology), let

$$C \;=\; \frac{1}{K}\sum_{k=1}^{K}\mathbf{1}[\text{customer field } k \text{ populated}] \;\in\; [0,1]$$

denote a continuous customer-completeness index, where the sum runs over the K logged customer-attribute fields outside the previous-flight family. Let $Y = \mathbf{1}[\text{previous-flight customer profile present}]$ be a binary indicator for the dropped customer family most strongly suspected of being proxied, defined over the previous-flight customer-attribute fields. By construction C and Y are computed from disjoint field sets, so that $r_{pb}(Z, C)$ and $V(Z, Y)$ probe independent slices of the customer namespace. Each of the three measures that follow relates Z to either Y (the risk ratio and Cramér's V) or C (the point-biserial correlation).

The risk ratio

$$\mathrm{RR} \;=\; \frac{P(Y = 1 \mid Z = 1)}{P(Y = 1 \mid Z = 0)}$$

compares how often a previous-flight profile is present when the flagged predictor is populated ($Z = 1$) against when it is not ($Z = 0$). It is the most direct single number, but it speaks only to the one dropped family encoded in Y. A value of RR = 1 means no association; values above one read directly as a fold-change (RR = 6 is a six-fold lift), and a value below one, which is not expected here, would indicate the opposite direction.

The point-biserial correlation between Z and the customer-completeness index C,

$$r_{pb} \;=\; \frac{M_1 - M_0}{s_C}\sqrt{\frac{n_1\,n_0}{n^2}},$$

is the Pearson correlation adapted to one binary and one continuous variable, where M_1 and M_0 are the mean completeness C in the $Z = 1$ and $Z = 0$ subgroups, s_C is the standard deviation of C, and n_1, n_0, n are the subgroup and total sample sizes. It widens the question from the single dropped family Y to the customer-feature space as a whole: does the flagged predictor's populated-ness move together with the average completeness of the other customer fields? It runs over $[-1, 1]$, with 0 meaning no linear association and positive values meaning the $Z = 1$ subgroup is the more complete one. As a rough scale, $|r_{pb}| \approx 0.1$ is small, 0.3 medium, and 0.5 large.

Cramér's V on the 2×2 contingency between Z and Y,

$$V \;=\; \sqrt{\frac{\chi^2}{n}}, \qquad \chi^2 \;=\; \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

rescales the chi-squared statistic to $[0, 1]$ so that it does not grow with sample size, where O_{ij} and E_{ij} are the observed and expected cell counts under independence (for a 2×2 table the general denominator reduces to n). This matters because at $n = 64{,}006$ the raw χ^2 and its p-value are uninformative: any non-zero association is significant by construction, so effect size, not significance, is what counts. Because V rests on different assumptions from RR and r_{pb}, agreement across all three guards against any single measure being misleading. It runs over $[0, 1]$, with 0 meaning independence and 1 a perfect relationship, and the same rough scale applies ($V \approx 0.1$ small, 0.3 medium, 0.5 large).

The three measures discriminate between three candidate explanations for a flagged predictor's high attribution, summarised in Table 4. A value-driven signal leaves Z unrelated to either C or Y, so all three measures sit near their null. A narrow artefact tied to one dropped field couples Z tightly to Y but only weakly to the broader pattern in C, so only RR moves materially. A broad missingness proxy for the dropped customer cluster couples Z to both Y and C, elevating all three measures together. The probe is therefore not a single hypothesis test but a small profile of effect sizes, read jointly against the predictor's business-logical story from the second step.

This probe is offered as a generalisable auditing step for CDH-XAI pipelines, complementing the RQ1 fidelity check: where RQ1 establishes that the explanations are faithful to the surrogate, the probe checks that the surrogate's attribution to a flagged predictor in turn tracks its value rather than the missingness signature of the platform's predictor-selection mechanism. The Pega ADM selection mechanism itself is treated as documented platform behaviour and is not re-derived here.

Table 4: Expected effect-size pattern under each candidate explanation for a flagged predictor's high attribution. The three measures are read jointly; the joint pattern, not any single value, identifies the underlying mechanism.

Candidate explanation	RR	r_{pb}	V
Value-driven signal	≈ 1	≈ 0	≈ 0
Narrow one-field missingness proxy	$\gg 1$	≈ 0	small
Broad missingness proxy (dropped cluster)	$\gg 1$	substantial	substantial

Chapter 3

Results

3.1 Surrogate Fidelity (RQ1)

RQ1 asks how accurately the CatBoost surrogate g approximates the Pega ADM black-box model f on the L5B15 variant. This section reports the candidate model comparison that motivates the architecture choice, the depth selection outcome, and the final fidelity of the selected surrogate against the five metrics defined in Section 2.5.

3.1.1 Candidate Model Comparison

Table 5 reports fidelity on the held-out 20% test set for all five candidate surrogates, evaluated jointly across the full metric suite.

Table 5: Surrogate architecture comparison on the L5B15 held-out test set (stratified 80/20 split). Five candidates are evaluated on rank-based fidelity (Spearman ρ, Kendall τ), pointwise accuracy (R^2, RMSE), and distributional alignment (KS statistic). Best value per metric in bold.

Model	R^2	RMSE	Spearman ρ	Kendall τ	KS statistic
Linear regression	0.469	0.0070	0.736	0.534	0.221
Naive Bayes	0.368	0.0077	0.862	0.658	0.268
Random Forest	0.761	0.0047	**0.934**	0.788	0.139
LightGBM	0.838	0.0039	0.924	0.779	**0.090**
CatBoost	**0.927**	**0.0026**	0.933	**0.798**	0.154

CatBoost leads convincingly on the pointwise metrics R^2 and RMSE, and on Kendall's τ, with Random Forest the closest competitor on Spearman ρ and LightGBM producing the lowest

KS statistic. The margins on those two metrics are small, however, while CatBoost's advantages on R^2 and RMSE are substantial across all tree-based candidates. The Gaussian Naive Bayes baseline is included as an architecture-matched reference, since the black-box model itself is a Naive Bayes classifier. It outperforms linear regression on rank-based metrics, but lags behind all tree-based candidates across the board. Two further considerations support the CatBoost selection beyond raw performance. First, its native categorical handling avoids the artificial ordering effects introduced by integer-based categorical encodings used in the other tree-based candidates. Second, its tree structure enables exact Shapley value computation via TreeSHAP, as described in Section 2.3. On the full metric suite, CatBoost is therefore the preferred surrogate architecture.

3.1.2 Depth Selection

Five-fold cross-validation over depths 1 to 15 is reported in Appendix A.2. The CV-maximum is reached at depth 12 and Breiman's 1-SD rule selects depth 10, whose mean ρ_S falls within one standard deviation of the CV maximum. The $\bar{\rho}_S$ curve plateaus across depths 10 to 13 before declining marginally at depths 14 and 15, confirming that the optimum lies well inside the grid and that the 1-SD selection is not an artefact of the search range.

3.1.3 Final Surrogate Fidelity

The CatBoost surrogate fitted at depth 10 on 80% of the L5B15 data achieves the fidelity reported in Table 5 on the held-out 20%. Across the primary rank-based criteria the surrogate reproduces the ordering of Pega's propensity scores with high fidelity, with τ_b confirming this agreement at the level of individual pairs. The pointwise fit is similarly strong, with R^2 indicating that the surrogate explains the large majority of variance in logged propensity scores. The moderate D_{KS} indicates that the surrogate captures the rank structure of f more faithfully than its absolute score distribution, which is an expected consequence of optimising depth selection on ρ_S rather than a distributional criterion and is acceptable given that rank order is the operationally relevant property of Pega's scoring. Together these results indicate that g provides a sufficiently faithful approximation of f to serve as a valid explanation target, satisfying the core modelling assumption established in Section 2.1.

3.2 Explanation Stability (RQ2)

RQ2 asks how consistently feature importance rankings are reproduced across temporal, route-based, and culture-based splits for the L5B15 variant. Stability is assessed using Spearman rank correlation ρ_S on the full feature ranking and Jaccard similarity at $k = 5$ and $k = 10$, as defined in Section 2.5. Results are reported for all three explanation methods across all five split comparisons in Table 6.

Table 6: Stability of feature importance rankings across L5B15 splits. ρ_S measures full-ranking consistency; J_5 and J_{10} measure top-k overlap. Route pair numbering follows Table 3.

	SHAP			LIME			Decision Tree		
Split	ρ_S	J_5	J_{10}	ρ_S	J_5	J_{10}	ρ_S	J_5	J_{10}
Temporal	0.996	1.000	1.000	0.989	1.000	0.818	0.855	0.667	0.667
Route pair 1	0.994	1.000	1.000	0.986	0.667	0.818	0.510	0.429	0.538
Route pair 2	0.990	1.000	0.818	0.858	1.000	0.818	0.584	0.250	0.667
Route pair 3	0.987	0.667	0.818	0.992	1.000	1.000	0.817	0.667	0.667
Culture (FR / NL)	0.958	0.667	1.000	0.949	1.000	0.667	0.666	0.429	0.667

3.2.1 SHAP Stability

SHAP produces the most stable rankings across all splits, with ρ_S ranging from 0.958 to 0.996 and full top-5 overlap on the temporal split and the route pairs. The culture split shows the largest deviation for SHAP, with $J_5 = 0.667$ indicating one feature difference in the top five between FR and NL, though the full-ranking correlation remains high. This high stability is structurally expected: SHAP values are computed from a single surrogate fitted once on the full training set, meaning any variation in rankings across splits reflects only genuine differences in how the surrogate weights features in different subpopulations, with no contribution from method refitting variance.

3.2.2 LIME Stability

LIME rankings are similarly stable across temporal and route splits, with ρ_S exceeding 0.986 in all cases except route pair 2 (ORY→OPO vs NCE→ORY), where ρ_S drops to 0.858. This is the only comparison with no shared origin or destination (Table 3), suggesting that LIME's local linear approximations are more sensitive to subgroup composition when the two groups share no structural route element. The culture split also shows reduced LIME stability relative to other splits, with $J_{10} = 0.667$ indicating three features differ in the top ten between the French and Dutch segments. Despite these cases, LIME's full-ranking stability is substantially higher than the decision tree across every split.

3.2.3 Decision Tree Stability

The decision tree is considerably less stable than SHAP and LIME across all split types. On route splits the instability is most pronounced: ρ_S reaches a minimum of 0.510 for route pair 1 (shared origin, different destination) and 0.584 for route pair 2 (different origin and destination), with J_5 as low as 0.250, indicating that only one of the five top-ranked features is shared between the two routes in that comparison. Route pair 3 (different origin, shared destination) is comparatively more stable at $\rho_S = 0.817$. The culture split produces the second largest instability, with $\rho_S = 0.667$ and $J_5 = 0.429$ indicating that the French and Dutch segments created substantially different decision tree structures. The temporal split is the most stable of the decision tree comparisons at $\rho_S = 0.855$, reflecting that the overall feature distribution shifts only moderately between the early and late observation windows.

To confirm that the observed decision tree instability reflects genuine subgroup sensitivity rather than random fitting variance, a bootstrap procedure is applied as described in Section 2.5: the decision tree is re-fitted ten times on the same data using different random seeds, and the mean pairwise ρ_S across seeds serves as a within-split baseline. Table 7 reports the results per split. For the temporal and culture comparisons the between-split ρ_S falls clearly below the within-split bootstrap mean, confirming that the observed instability exceeds what fitting noise alone would produce. The culture split shows the largest gap, indicating that the French and Dutch segments produce genuinely different tree structures rather than random variation across fits.

For the route pairs the results are more nuanced. Route pairs 1 and 2 show substantial gaps between the bootstrap baseline and the between-split ρ_S, confirming genuine subgroup-driven instability. Route pair 3, however, shows a negative gap: the between-split ρ_S (0.817) slightly exceeds the mean within-split ρ_S (0.802), meaning the observed variation between NCE→ORY and TLS→ORY is fully within the range of normal fitting noise. The decision tree is not meaningfully unstable across this pair. This result contextualises the stability metric in Table 6: while $\rho_S = 0.817$ is lower than SHAP or LIME on the same pair, it does not reflect a genuine subgroup difference and should not be interpreted as evidence of explanation instability.

Table 7: Bootstrap validation of decision tree instability. Within-split ρ_S is the mean pairwise Spearman correlation across ten re-fits on the same data; between-split ρ_S is the observed stability metric from Table 6. A positive gap confirms that between-split instability exceeds what fitting noise alone can explain; a negative or near-zero gap indicates that the observed variation falls within the range of random fitting variance.

Split	Within-split ρ_S	Between-split ρ_S	Gap
Temporal	0.935	0.855	0.080
Route pair 1 (ORY→NCE vs ORY→OPO)	0.824	0.510	0.314
Route pair 2 (ORY→OPO vs NCE→ORY)	0.789	0.584	0.206
Route pair 3 (NCE→ORY vs TLS→ORY)	0.802	0.817	−0.015
Culture (FR / NL)	0.977	0.667	0.310

3.2.4 Stability at the Feature-Group Level

Figure 2 shows the normalised feature importance scores across all splits for each explanation method, with features ordered by mean importance across all splits and methods combined. The SHAP and LIME panels present a smooth, consistent colour gradient across splits: the same features dominate the top rows uniformly, with importance decaying gradually down the ranking and the pattern shifting only marginally between splits. The decision tree panel presents a different picture: high-contrast patches appear in different rows across splits, reflecting the tree's tendency to concentrate importance on whichever single feature is most discriminative in a given subgroup rather than distributing it across the full feature set. This pattern is consistent with the instability gap reported in Table 6.

Two levels of stability emerge from this analysis, and the distinction matters for everything that follows. At the level of individual feature ranks, the methods diverge: SHAP and LIME are nearly invariant across splits, the decision tree is materially less stable, and the three methods only partially agree on the precise order of features within the top of the ranking. At the level of feature *groups*, by contrast, all three methods converge on the same broad attribution structure across every split tested. The top-ranked features belong consistently to two feature groups across all splits and methods, contact history features capturing the recency and frequency of prior outbound communications and offer-level scoring parameters, with booking context features playing a secondary but stable role; the decision tree's within-group reshuffling does not propagate to those group-level shares. This second result is the load-bearing finding for any downstream content analysis: irrespective of which method one reads or which split one looks at, the L5B15 explanations are stable enough at the group level to support a substantive reading of which feature groups the production model scores on, which is the question §3.5 takes up.

3.3 Instability Attribution (RQ3)

The stability values ρ_{DT}, ρ_{SHAP}, and ρ_{LIME} underlying the decomposition below are reported in Table 6. Table 8 decomposes the observed instability into its component sources following the classification rule in Table 2: For L5B15 the temporal split is classified as Both, and all four subgroup splits as Explainer.

The temporal split is the only case where external conditions are significant: both the propensity distribution and the model snapshot pool shift substantially between the early and late window. Despite this, all three explanation methods remain relatively stable, with SHAP and LIME showing near-perfect rank preservation and the decision tree showing modest instability. The DT refit gap of the temporal split is the smallest across splits, indicating that when the underlying system shifts, the dominant features driving propensity scores remain largely the same regardless of explanation method. The feature importance structure is robust to external variation.

The four subgroup splits tell the opposite story. Both external diagnostics fall below their re-

Table 8: L5B15 instability attribution. δ_d is the Kolmogorov–Smirnov statistic on the propensity distribution; flagged if ≥ 0.10. δ_m is the Jaccard overlap on the set of model versions active in each sub-population; flagged if ≤ 0.10. $\delta_e^{\mathrm{DT}} = \rho_{\mathrm{SHAP}} - \rho_{\mathrm{DT}}$ and $\delta_e^{\mathrm{LIME}} = \rho_{\mathrm{SHAP}} - \rho_{\mathrm{LIME}}$ are the refit and sampling sensitivity gaps respectively; both are descriptive and not used for source classification. Source assignment follows Table 2.

Split	δ_d	δ_m	δ_e^{DT}	δ_e^{LIME}	Source
L5B15 temporal	0.356	0.002	0.141	0.007	Both
Route pair 1 (ORY→NCE vs ORY→OPO)	0.099	0.481	0.484	0.008	Explainer
Route pair 2 (ORY→OPO vs NCE→ORY)	0.094	0.547	0.406	0.132	Explainer
Route pair 3 (NCE→ORY vs TLS→ORY)	0.086	0.525	0.171	-0.005	Explainer
Culture (fr-FR vs nl-NL)	0.081	0.595	0.291	0.009	Explainer

spective thresholds across all four splits: propensity distributions are stable and model snapshot overlap is high, confirming that these splits isolate the explanation methods from system-level changes. Yet the decision tree shows substantial ranking instability, with refit gaps ranging from 0.171 to 0.484, while SHAP rankings are near-perfect across all four splits. LIME tracks SHAP closely in three of the four splits, with a near-zero sampling sensitivity gap. The exception is the ORY→OPO vs NCE→ORY comparison, where LIME's sampling sensitivity rises to 0.132, suggesting that local perturbation sampling is more sensitive to the specific subgroup composition of that route pair. The within-route bootstrap validation confirms that the DT instability observed here reflects genuine subgroup sensitivity rather than random fitting noise: for three of the four subgroup splits the between-split ρ_{DT} falls more than 0.10 below the within-sample mean, well exceeding the noise floor established by ten resamples on identical data (Appendix A.7).

Taken together, these results answer RQ3 directly: under controlled conditions where external sources of variation are absent, the dominant source of observed instability is the explanation method itself. The decision tree is consistently the least stable method, driven by refitting variance rather than genuine model or data change. SHAP is the most stable across all splits, with LIME tracking it closely. Under large external shifts the decision tree continues to diverge from SHAP, although the magnitude of that divergence is smaller on the L5B15 temporal split ($\delta_e^{\mathrm{DT}} = 0.141$) than on its subgroup splits (0.171–0.484); SHAP and LIME continue to track each other closely throughout.

3.4 Cross-Offer Replication (RQ4)

RQ4 asks whether the fidelity and stability findings observed for the primary L5B15 luggage offer replicate across three further offer variants: CLUG (generic luggage), BookingDotCom (third-party accommodation), and Cartrawler (third-party car rental). The same surrogate-and-attribution pipeline is applied to each variant, with routes pinned to the L5B15 set for direct comparability and a culture split (French versus Dutch) added for each variant. Surrogate fidelity per variant is reported in Table A.4; the per-variant instability attribution is reported in Table 9.

Table 9: RQ4 replication: attribution summary for CLUG, BookingDotCom, and Cartrawler, grouped by split. Routes are pinned to the L5B15 set (ORY→NCE, ORY→OPO, NCE→ORY, TLS→ORY) for direct comparability. Metric definitions and thresholds match Table 8: δ_d from the Kolmogorov–Smirnov statistic on propensity, flagged if ≥ 0.10; δ_m from Jaccard overlap on the set of active model versions, flagged if ≤ 0.10. δ_e^{DT} and δ_e^{LIME} are descriptive and not used for source classification.

Variant	δ_d	δ_m	δ_e^{DT}	δ_e^{LIME}	Source
Temporal splits					
CLUG	0.390	0.002	0.421	0.005	Both
BookingDotCom	0.246	0.002	0.166	0.006	Both
Cartrawler	0.721	0.002	0.373	0.025	Both
Route pair 1 (ORY→NCE vs ORY→OPO)					
CLUG	0.052	0.406	0.363	0.015	Explainer
BookingDotCom	0.116	0.416	0.711	-0.036	Dist. shift
Cartrawler	0.079	0.404	0.512	0.009	Explainer
Route pair 2 (ORY→OPO vs NCE→ORY)					
CLUG	0.116	0.404	0.266	0.197	Dist. shift
BookingDotCom	0.224	0.352	0.471	-0.024	Dist. shift
Cartrawler	0.059	0.361	0.757	0.367	Explainer
Route pair 3 (NCE→ORY vs TLS→ORY)					
CLUG	0.069	0.426	0.490	0.015	Explainer
BookingDotCom	0.062	0.396	0.333	-0.010	Explainer
Cartrawler	0.062	0.376	0.571	0.010	Explainer
Culture (fr-FR vs nl-NL)					
CLUG	0.114	0.464	0.383	-0.017	Dist. shift
BookingDotCom	0.718	0.529	0.054	-0.060	Dist. shift
Cartrawler	0.689	0.564	0.177	0.352	Dist. shift

3.4.1 Surrogate Fidelity Across Variants

Surrogate fidelity is reported in table A.4.The CatBoost surrogate achieves high pointwise fidelity on both luggage variants ($R^2 = 0.927$ on L5B15, 0.954 on CLUG) and on Cartrawler ($R^2 = 0.912$); BookingDotCom is lower at $R^2 = 0.793$ despite its large sample, indicating that some BookingDotCom variance is not captured at the chosen surrogate depth. Rank fidelity (Spearman ρ) is comparably strong on three of the four variants (0.92–0.95) but drops to 0.615 on Cartrawler, meaning the surrogate reproduces Cartrawler's propensity ordering noticeably less faithfully than the other variants. This caveat is carried forward when reading Cartrawler's downstream attribution numbers: explanation diagnostics inherit the surrogate's approximation error and Cartrawler-specific results should be interpreted with the lower rank-fidelity baseline in mind.

3.4.2 Temporal Split: Full Replication

The temporal pattern observed for L5B15 replicates across all three additional variants. Every variant is classified as Both, indicating significant propensity drift ($\delta_d \geq 0.10$) accompanied by near-complete model-version turnover ($J_v < 0.005$) between the early and late observation windows. The magnitude of propensity drift varies substantially across variants, ranging from $\delta_d = 0.246$ on BookingDotCom to 0.721 on Cartrawler (with L5B15 at 0.356 and CLUG at 0.390), yet the classification is identical because the model-version overlap is uniformly near zero. This is the expected consequence of Pega ADM's continuous online learning: any time window long enough to span meaningful population dynamics also spans many micro-updates of the underlying Naive Bayes, so temporal splits structurally combine data drift and snapshot turnover regardless of which offer is being scored. The DT refit-sensitivity gap δ_e^{DT} is substantial across the temporal split for every variant (0.141–0.421) and is notably larger for CLUG and Cartrawler than for L5B15 itself, so the decision tree continues to diverge from SHAP under externally-driven shifts and not only under the controlled subgroup conditions of RQ3. By contrast, LIME sampling sensitivity stays near zero in all four variants (largest $\delta_e^{\mathrm{LIME}} = 0.025$ on Cartrawler), meaning LIME continues to track SHAP closely throughout. The pattern that emerges is consistent across split types: SHAP and LIME agree on the top-feature ranking whether the underlying instability is externally-driven or controlled, while the decision tree disagrees with both regardless of which source is driving the shift.

3.4.3 Route Subgroups: Largely Replicates with Two Reclassifications

Route subgroup splits replicate the L5B15 Explainer pattern across most cells, with reclassifications to Dist. shift on cells where δ_d marginally crosses the threshold. Route pair 3 (NCE→ORY vs TLS→ORY) is classified as Explainer for every variant including L5B15, with δ_e^{DT} between 0.171 and 0.571 in every cell. Route pair 1 (ORY→NCE vs ORY→OPO) is Explainer for L5B15, CLUG, and Cartrawler; BookingDotCom alone flips to Dist. shift because its δ_d marginally crosses the threshold (0.116), even though its δ_e^{DT} on this row is the largest in the entire table at 0.711. Route pair 2 (ORY→OPO vs NCE→ORY) is Explainer for L5B15 and Cartrawler, and Dist. shift for CLUG ($\delta_d = 0.116$) and BookingDotCom ($\delta_d = 0.224$).

The qualitative interpretation of the RQ3 finding holds even on the reclassified rows: in every route pair and every variant, δ_e^{DT} remains substantial (0.266–0.757), so the decision tree continues to contribute to ranking instability whether or not the external diagnostic crosses the threshold. The reclassifications to Dist. shift therefore do not reverse the Explainer story; they indicate that in some non-L5B15 route pairs the external trigger is also present alongside the explainer gap. LIME sampling sensitivity is small or slightly negative across most route cells, consistent with the L5B15 pattern, with one notable exception: Cartrawler route pair 2 shows $\delta_e^{\mathrm{LIME}} = 0.367$, indicating that LIME's local perturbation sampling is unusually unstable for that specific subgroup of Cartrawler. This figure should be read with some caution, since Cartrawler has the weakest surrogate rank-fidelity of the four variants (Section 3.4), so its downstream attribution diagnostics rest on a less reliable base.

3.4.4 Culture Split: A Divergent Finding

The culture split is the one place where the L5B15 finding does not replicate. L5B15 is classified as Explainer because its culture-driven propensity shift sits just below the threshold ($\delta_d = 0.081$). All three replication variants flip to Dist. shift: CLUG shows a modest above-threshold shift ($\delta_d = 0.114$), and BookingDotCom and Cartrawler show very large shifts ($\delta_d = 0.718$ and 0.689 respectively). The substantive reading is that customer culture (French versus Dutch) maps onto materially different propensity distributions for the cross-sell and third-party offers but not for the luggage upsell. That is, the Pega model has learned a stronger culture-conditional response pattern in BookingDotCom and Cartrawler than in the luggage offers, plausibly because accommodation and car-rental preferences vary more across European customer segments than basic luggage add-on uptake does.

The two third-party variants also produce instructive explainer-gap patterns under culture. BookingDotCom's δ_e^{DT} collapses to 0.054 on the culture split, the lowest in the entire table, meaning that despite a very large external shift the decision tree's ranking is comparatively stable across cultures for that variant. Cartrawler shows the opposite pattern, with $\delta_e^{\mathrm{LIME}} = 0.352$ alongside its large δ_d, indicating that LIME's local sampling becomes unstable in the same comparison that also exhibits substantial external drift. These two cells illustrate that even when external sources fire, the relative contribution of refit and sampling sensitivity can vary substantially across product contexts.

3.4.5 Summary of Replication Evidence

The replication evidence supports the RQ3 conclusion with one important caveat. The temporal pattern of externally-driven instability with moderate DT contribution replicates fully across all four variants. The route pattern of explainer-dominant instability under controlled same-window conditions replicates qualitatively across all variants, with three rows reclassified to Dist. shift on borderline δ_d values; the underlying DT-versus-SHAP gap remains the larger numerical contributor in those cells. The culture pattern does not replicate: L5B15 culture instability is classifiable as Explainer-driven only because its δ_d sits just below threshold, whereas the three other variants exhibit genuine culture-induced propensity shifts large enough to dominate the attribution. This divergence is best read as a property of the underlying Pega model rather than of the evaluation pipeline: customer culture has a stronger effect on third-party and generic-luggage propensity than on the L5B15 weight-specific luggage offer. The methodological findings therefore generalise, while the substantive role of any one feature dimension in driving observed instability is product-context-specific.

3.5 Feature Drivers and Business Implications (RQ5)

§3.2 established that the three explanation methods converge on the same L5B15 attribution structure at the level of feature groups across every split tested, even where their individual feature ranks diverge, and §3.3 attributed the residual individual-rank drift to decision-tree refit variance (δ_e^{DT}) rather than to underlying model or data change. Together those two results allow us to read group-level attribution shares as a genuine description of what the production model has learned to score on. They also dictate how the three methods should be weighed here. Because RQ2 and RQ3 found SHAP and LIME to be the reliable, stable methods and the decision tree to be the unstable one, this section leads with SHAP and LIME and treats the decision tree as a secondary check rather than an equal partner: the per-feature ranking is ordered by the SHAP/LIME consensus alone, and the decision-tree share is reported alongside but not averaged in. Where the decision tree nonetheless agrees on direction, that agreement strengthens the reading precisely because it comes from the method most prone to disagree. This section takes up that opportunity and asks the content question: which features drive the L5B15 propensity score, and what the resulting attribution structure means for the marketing and governance stakeholders who consume those scores.

3.5.1 What the Model Scores On

Table 10 reports the per-feature importance share for each method, ordered by the SHAP/LIME consensus rank. The offer-identity scoring parameter carries the single largest share in every method, as expected for a feature that simply encodes which product is being scored. Setting that aside, the highest-ranked predictors are overwhelmingly contact-history features: the number of days since the last outbound email and the historical count of prior outbound contacts. These are recency and frequency signals, the two quantities that together define contact fatigue. The decision tree concentrates an especially large share on a single such feature, the historical outbound-email count, consistent with its tendency to split early on whichever recency or frequency variable is most discriminative. This concentration is exactly the behaviour RQ3 traced to the tree's refit variance, and it is why the ordering here is governed by the SHAP/LIME consensus rather than by an average that would let the tree's single-feature spike distort the ranking; the decision-tree column is retained so that its broad agreement with that ordering can still be read off.

3.5.2 Recency and Fatigue, Not Customer Identity

Aggregated to the feature-group level (Table 11), all three methods agree on the same ordering: contact-history features are the largest group and flight-level features the second. They disagree on the size of the gap. Contact history accounts for roughly a third to a half of total importance

Table 10: L5B15 feature-importance shares across the three explanation methods, ordered by the SHAP/LIME consensus rank (the mean of the SHAP and LIME within-method ranks). The ordering uses only the two methods that RQ2 and RQ3 found reliable; the decision-tree share is reported as a corroborating column but does not drive the ordering. Each method's importances are normalised to sum to 100%. Group tags: CH = contact history, FD = flight data, BC = booking context, SP = scoring parameter.

Feature	Grp	SHAP%	LIME%	DT%	SL rank
Param.BundleName	SP	15.7	19.0	21.5	1.0
Email.Out.Pending.DaysSinceLast	CH	8.5	5.5	2.2	3.5
SeatNumber	FD	6.9	10.3	2.3	3.5
IsStaffStandBy	FD	6.4	7.1	0.0	4.5
Email.Out.Delivered.DaysSinceLast	CH	7.1	4.0	10.1	5.5
BookingMonth	BC	6.1	6.7	1.6	6.0
Email.Out.Pending.HistCount	CH	6.0	4.8	34.5	7.5
Push.Out.Pending.LastGroup	CH	8.0	3.1	0.2	8.5
Email.Out.Pending.LastGroup	CH	6.2	3.3	0.0	10.0
FlightInboundArrival	BC	4.9	3.8	1.1	10.0
Language	FD	4.0	3.8	0.1	10.0
Email.In.Clicked.DaysSinceLast	CH	2.9	3.5	17.0	11.5
BookerGender	BC	2.4	3.5	1.3	13.5
Push.Out.Pending.DaysSinceLast	CH	2.8	2.5	2.2	14.0
DestinationAirport	FD	0.6	3.9	0.0	15.0
Event.Out.RealTimeEvent.HistCount	CH	2.8	1.9	0.0	16.5
Email.Out.Delivered.LastGroup	CH	2.0	2.3	0.0	16.5
AirlineCodeIATA	FD	0.9	2.4	0.0	18.0
Email.In.Pending.DaysSinceLast	CH	1.9	1.8	1.0	18.5
Journey	FD	0.9	1.9	1.2	19.5
Push.Out.Pending.HistCount	CH	1.6	1.5	3.6	20.0
CultureCode	BC	1.4	1.8	0.1	20.0
FlightNumberOperatorIATA	FD	0.1	1.3	0.0	23.0

under LIME and SHAP and for the clear majority (around 70%) under the decision tree, so the exact share is best read as a range rather than a single figure. The direction of the result, that contact history dominates, holds regardless of which method is used. Booking context and the scoring parameter make up the remainder. Customer-level attributes, demographics and loyalty profile, contribute nothing, because no customer attribute survives Pega's active-predictor selection for L5B15 in the first place (Section 2.2).

The commercial reading is direct. For this offer, the propensity model is, in effect, scoring how recently and how often the customer has been contacted, together with the context of the booked trip, rather than who the customer is. Loyalty tier, demographics, and historical value, the attributes marketing teams most often assume drive personalisation, play no role in the L5B15 score because they were never retained as predictors. What this points to is contact strategy. Because recency and contact-frequency carry most of the score, the part of the input the model reacts to most is the cadence of the outbound programme itself. Two cautions apply before turning this into a lever. First, the importance figures show how much each feature matters, not

Table 11: L5B15 feature-importance aggregated by feature group, as a percentage of each method's total importance. Contact-history (recency and frequency of prior outbound contact) and flight-level features dominate across all three methods; customer-level attributes contribute nothing because none survive Pega's active-predictor selection for L5B15.

Feature group	# feat	SHAP%	LIME%	DT%
Contact history	11	49.8	34.4	70.8
Flight data	7	19.7	30.7	3.6
Booking context	4	14.8	15.8	4.1
Scoring parameter	1	15.7	19.0	21.5
Customer attributes	0	0.0	0.0	0.0

in which direction it pushes the score, so they do not by themselves say whether more or less contact raises propensity. Second, contact recency is also correlated with whether a customer is a recognised returning traveller, so part of what looks like a contact-cadence effect may really be a returning-customer effect. Even with those caveats, contact-cadence management is a more plausible lever here than richer customer segmentation. Segmentation cannot help at all, because no customer attribute was retained as a predictor in the first place. This also tempers a common governance expectation: an audit asking "which customer characteristics drive this offer?" would find that, for L5B15, the honest answer is essentially none.

3.5.3 A Top Predictor Encoding Missingness, Not Value

One flight-group predictor warrants separate scrutiny. The staff-standby flag, which marks bookings issued on a staff standby ticket, ranks among the most important features under both SHAP and LIME (Table 10), which is commercially counter-intuitive: staff standby travel should have little to do with a passenger's propensity to buy a luggage add-on. Inspection of the raw field explains the anomaly. The flag is almost never true; what varies across records is whether the field is populated at all. Table 12 shows that records for which the field is populated are far more likely to carry a previous-flight customer profile than records for which it is blank (20.3% versus 3.0%, a 6.8× difference; point-biserial $r = 0.25$ and Cramér's $V = 0.25$ with the presence of a previous-flight profile, both $p \approx 0$).

The predictor is therefore acting as a proxy for record completeness rather than for staff-standby status: its populated-ness co-varies with whether the customer is a recognised returning traveller with historical profile data attached. This is the visible downstream symptom of Pega's predictor selection. Because the platform deactivates sparse customer-level predictors and retains only the strongest of any group of correlated predictors, the missingness pattern those predictors would have carried does not disappear; it re-emerges, encoded in the presence pattern of a surviving feature that happens to correlate with data availability. This is a single, post-hoc case rather than a systematic survey of every predictor, but the governance implication is concrete

Table 12: Test of the missingness-proxy hypothesis for the IsStaffStandBy flag on L5B15. The flag is never true in the sample, and what varies across records is whether the field is populated at all; where it is populated, customer-profile data is far more likely to be present, indicating that the surrogate (and the underlying model) keys on the field's presence as a proxy for record completeness rather than on staff/standby status itself.

Quantity	Value
n (L5B15 records)	64,006
IsStaffStandBy populated rate	0.582
records with IsStaffStandBy = true (overall and among populated)	0
customer-attribute fields used for the completeness index	75
previous-flight customer-attribute fields	33
mean customer-completeness — populated	0.755
mean customer-completeness — blank/null	0.714
P(previous-flight profile — populated)	0.203
P(previous-flight profile — blank/null)	0.030
risk ratio	6.8x
point-biserial r (populated vs. completeness)	0.251 ($p < 0.001$)
Cramér's V (populated vs. previous-flight profile)	0.253 ($p < 0.001$)

and likely reaches beyond it: in a continuously self-selecting feature space, a feature's measured importance can reflect which records have data rather than what the data says, and a predictor that an audit would flag as nonsensical may in fact be the model's handle on an unmodelled missingness structure. Detecting this requires checking high-importance predictors against their own populated-ness, a cheap diagnostic that this case study shows is worth running before attribution results are presented to stakeholders.

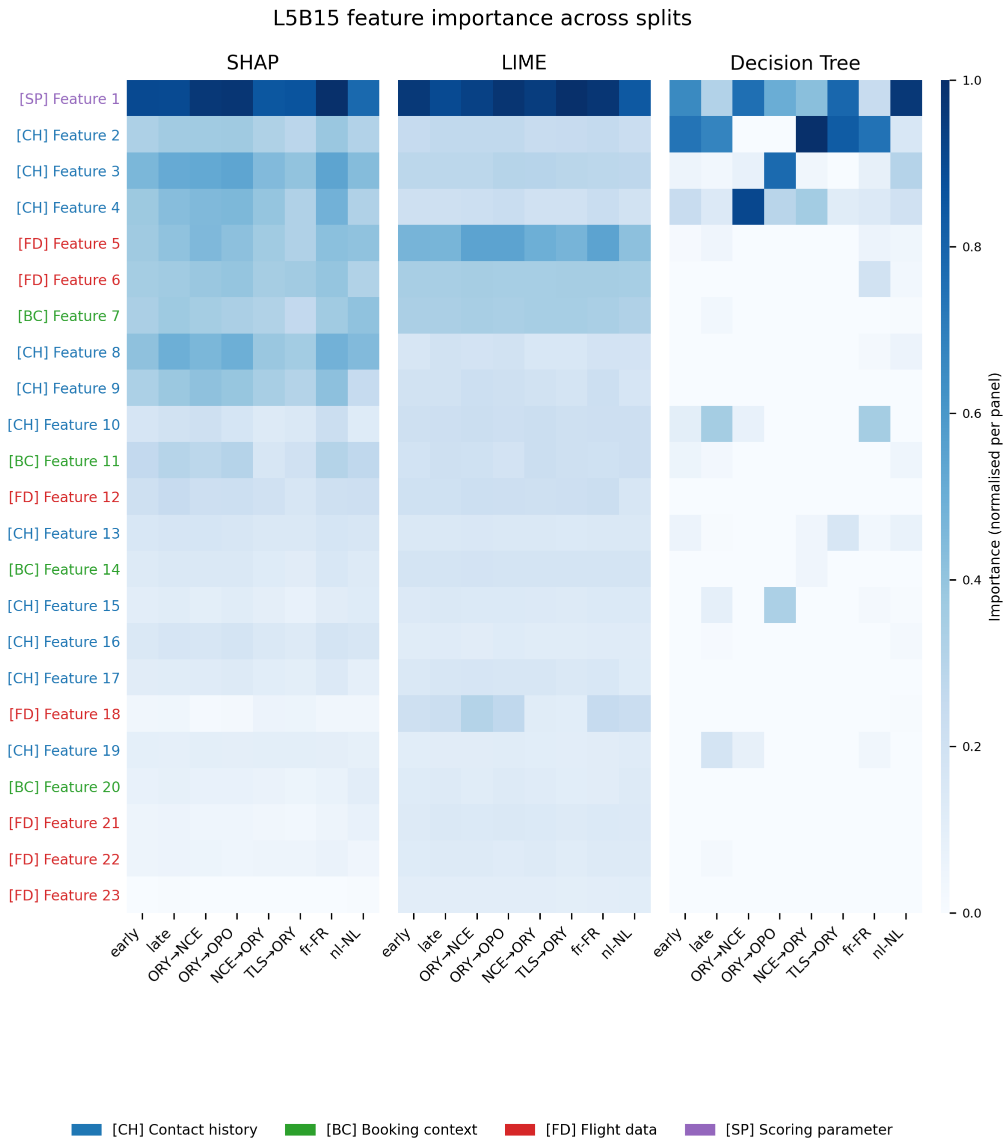

Figure 2: Normalised feature importance across L5B15 splits for SHAP, LIME, and the decision tree surrogate. Importance scores within each panel are normalised to $[0, 1]$ by dividing by the panel maximum. Features are ordered by mean importance across all splits and methods, most important at top. Feature names are anonymised; features belong to four groups: contact history, booking context, flight data, and scoring parameters.

Chapter 4

Discussion

4.1 Summary of Key Findings

Across the five research questions the empirical picture is internally consistent. A CatBoost surrogate reproduces Pega ADM's logged propensity scores with high rank fidelity on three of the four offer variants studied, providing a sufficient base for downstream attribution. Built on that surrogate, SHAP and LIME produce highly stable feature rankings across temporal, route, and customer-culture splits, while a shallow decision tree surrogate fitted directly on the propensities produces rankings that vary substantially between subgroups. The instability-attribution diagnostic shows that under controlled same-window conditions, where neither the propensity distribution nor the active model snapshots shift meaningfully, the dominant source of observed ranking instability is the explanation method itself. In other words, the explanation a stakeholder ends up with depends partly on which method was used to produce it, which is an uncomfortable property in a business setting where that explanation is meant to support a decision. The pattern replicates across three further offer variants on temporal and route splits. The one place L5B15 diverges from the replication set is the customer-culture split: for the third-party offers (accommodation and car rental), the propensity distribution differs much more between French and Dutch customers than it does for the luggage offers. A plausible reason is that preferences for accommodation and car rental vary more across European markets than the decision to buy extra luggage does, so the model has learned a stronger culture-dependent response for those offers.

Across explanation methods, SHAP is the most reliable in this setting. LIME tracks SHAP closely with one exception linked to its local perturbation sampling. The decision tree surrogate, despite being interpretable in the strictest end-to-end sense, is the least stable across populations even when the underlying model has not changed.

4.2 Interpretation of Findings by Research Question

4.2.1 Surrogate Fidelity (RQ1)

The fidelity result is unsurprising in itself: a flexible gradient-boosted ensemble trained directly on logged scores would be expected to reproduce those scores reasonably well. What matters here is not the level itself but its consequence for the rest of the pipeline. Because SHAP and LIME operate on the surrogate g, every attribution they produce is downstream of g's approximation of f. The decision tree surrogate, fitted directly on p, faces a single layer of approximation rather than two. Strong rank fidelity on three of four variants therefore matters less as a result in itself than as the precondition that makes the SHAP and LIME attribution claims trustworthy in those settings.

The Cartrawler exception illustrates the practical importance of reporting rank-based fidelity alongside R^2. Cartrawler's pointwise fit is comparable to the other variants, yet its Spearman rank correlation is markedly lower. The surrogate captures Cartrawler's propensity scale but reproduces its ordering less faithfully. For a system that consumes propensities through prioritisation rather than calibration, the rank disagreement is the more consequential one. Cartrawler's downstream attribution results are therefore read with that fidelity caveat in mind, and given less weight than those of the other variants.

The candidate-model comparison underlying the architecture choice also has interpretive value. A Gaussian Naive Bayes baseline, architecture-matched to Pega's internal model, lags behind every tree-based candidate on the fidelity suite. This matches a straightforward intuition: the best surrogate of a black-box model need not belong to the same model class as the model itself, particularly when feature interactions and high-cardinality categoricals are involved. The point is consistent with general treatments of interpretable surrogate models (Molnar, 2025), but it is shown here directly through the candidate comparison rather than assumed. A flexible surrogate produces more faithful approximations to the model output, even when the underlying model is itself a simpler classifier. In the present setting, two factors plausibly drive this gap. First, Pega ADM's online learning means the model keeps changing over the observation window. Pooled across that window, the logged scores come from many slightly different Naive Bayes snapshots rather than from one fixed classifier, so the overall input-output mapping behaves like a mixture of models. A flexible surrogate can fit this mixture more closely than any single Naive Bayes model could. Second, Pega bins its categorical predictors in a way that discards information, whereas CatBoost encodes the same categorical features natively, using statistics of the target, and so preserves more of it.

4.2.2 Explanation Stability (RQ2)

The stability results speak to a concern raised in the XAI evaluation literature, that explanations of the same model can diverge across populations even when those populations are drawn from the same data-generating process (Ribeiro et al., 2016; Vilone & Longo, 2020). The contribution here is to show how that divergence behaves across three explanation families under deliberately controlled conditions.

SHAP's high stability is structurally expected. Because a single CatBoost surrogate is fitted once on the full training set and applied to both sub-populations of every split, TreeSHAP's deterministic computation leaves no room for refitting variance. Any observed ranking shift reflects only how the surrogate weighs features differently in different sub-populations. That this shift remains small across temporal, route, and culture splits indicates that the surrogate's attribution structure is itself stable, not just that the method is deterministic.

LIME's close tracking of SHAP is more notable than it might appear at first reading. LIME shares the fixed-surrogate property with SHAP but introduces an additional source of variance through its perturbation-based local linear fits. That LIME nevertheless matches SHAP's stability on most splits suggests that the sampling variance is modest in practice for this dataset and feature space. The exception on the route pair with neither shared origin nor destination shows that this is not guaranteed: when two sub-populations share no structural element, local linear approximations can shift enough to move global aggregates.

The decision tree's instability is the central finding of RQ2. A shallow tree concentrates importance on whichever single feature is most discriminative in a given subgroup, rather than distributing it across the active feature set; small changes in subgroup composition can therefore reshuffle the top of the ranking even when the model is not asked to change. This is a property of tree-based importance more than of decision trees as such, and it has been documented in the broader interpretable-machine-learning literature (Molnar, 2025), but the magnitude observed here is substantial. The bootstrap validation rules out the alternative explanation that this is fitting noise: on three of four subgroup splits the between-split disagreement exceeds within-split disagreement by margins well outside what re-fitting on the same data can produce.

For a production decisioning system the operational implication is direct. A practitioner who inspects per-subgroup decision-tree explanations to understand why a model behaves differently on different customer segments may attribute that difference to the model itself, when in fact the underlying scoring function has barely moved and the difference lies in how the explanation method aggregates importance over the subgroup. Stability is therefore not only a property of the model being explained. It also depends on the explanation method, on the surrogate when one is used, and on how both interact with the evaluation data.

4.2.3 Instability Attribution (RQ3)

The instability-attribution diagnostic addresses a question that is often raised but rarely answered directly in the XAI evaluation literature (Molnar, 2025; Vilone & Longo, 2020): when explanations of a deployed model disagree across populations, where does the disagreement come from? The three-source decomposition, distribution shift δ_d, model churn δ_m, and explainer sensitivity δ_e, separates causes that are commonly conflated and provides a structured basis for assigning observed instability to its dominant source.

The L5B15 temporal split is the only case where both external sources fire simultaneously: the propensity distribution shifts substantially across the observation window, and model-snapshot overlap collapses to near zero. This is not a flaw of the split design but a structural property of continuous online learning. Any time window long enough to span meaningful population dynamics will also span many micro-updates of the underlying Naive Bayes. The temporal split therefore mixes data drift and snapshot turnover by construction, and a clean separation between the two in the temporal case is not available from observational logs alone.

The same-window subgroup splits, route and culture, are the test cases where the diagnostic does most of its work. Because those splits share an observation window, both δ_d and δ_m are constrained to be small by design, so any remaining ranking instability isolates the contribution of the explanation method. On every same-window split for L5B15 the decision tree's refit gap is substantial, SHAP rankings remain near-identical, and LIME tracks SHAP within sampling tolerance. The conclusion is therefore not that decision-tree-based explanations are inherently unreliable, but rather that under operationally realistic conditions where data and model are approximately stable, the choice of explanation method materially affects which features appear to drive scoring.

It is worth being clear that the diagnostic does not claim to pin down causes exactly. The three sources are estimated from effect-size statistics rather than isolated by intervention, and δ_d and δ_m are themselves correlated, since the same incoming response data that shifts the feature distribution also drives the online updates. The protocol is best read as an indicative decomposition that separates sources usually lumped into a single instability figure, not as a precise account of why a particular ranking moved.

4.2.4 Cross-Offer Replication (RQ4)

The replication evidence has both methodological and substantive readings. Methodologically, the protocol generalises: applying the same surrogate fitting, fidelity assessment, and attribution diagnostic to three further offer variants reproduces the qualitative pattern from L5B15 in nearly every cell. On temporal splits the instability is classified as jointly data- and model-driven (Both) across all four variants. On route splits it is classified predominantly as explainer-driven, with a few cells reclassified to distribution shift on borderline values of δ_d, and the decision tree's refit gap remains the largest numerical contributor in route cells even when the formal label flips. A practitioner applying this diagnostic to a new offer in the same CDH environment should expect to recover comparable patterns.

The picture for specific features is more nuanced, and in some ways more interesting. The culture split is the one place where L5B15 does not replicate, but the divergence is not a failure of the protocol. L5B15's customer-culture propensity shift sits just below the δ_d threshold, while the three replication variants cross it. For BookingDotCom and Cartrawler the magnitude is large; for CLUG it is modest. The interpretation is that customer culture, in this study French versus Dutch, maps onto materially different propensity distributions for the cross-sell and third-party offers but not for the L5B15 weight-specific luggage upsell. This is best read as a property of the underlying Pega model and the data it has accumulated on each offer, rather than of the evaluation pipeline.

Including offers from different commercial categories, rather than only further luggage variants, is what makes this distinction visible: the evaluation method behaves the same way across all of them, while the specific feature that drives instability depends on the offer. A set of only closely related luggage offers would still have tested the method, but would have shown less about how the diagnostic behaves when offers share most of their feature space yet differ in how customers respond to them.

4.2.5 Feature Drivers (RQ5)

The feature-level findings are, for a practitioner audience, the most directly consequential of the thesis. They are also where the validation work of RQ1–RQ4 pays off. Because SHAP and LIME were shown to represent the model faithfully and stably, their attributions can be read as a genuine window onto what the production model has learned, rather than as an artefact of the explanation method. In a well-specified model that window is normally used the way marketing teams want it: to see which signals actually drive uptake and to motivate the treatments and offers built on top of them. In this case study the same window does something less comfortable but equally useful, in that it mostly exposes gaps in the model rather than handing marketing a ready lever. Two results stand out, and both are gaps of this kind. The first is that the L5B15 propensity score is driven by contact recency and frequency and by the context of the booked trip, and not at all by customer identity. This is not a quirk of attribution but a structural consequence of how the model was built: Pega's predictor selection retained no customer-level attribute for this offer, so demographics and loyalty profile cannot drive the score regardless of which explanation method is used. The reading that the model scores contact fatigue more than it scores the customer is therefore robust to the explainer-stability concerns raised in RQ2 and RQ3, because it holds at the feature-group level where all three methods agree.

The second result is that a feature's importance can be an artefact of the data-collection process rather than a behavioural signal. The staff-standby predictor is influential not because staff-standby status predicts luggage purchase but because the field's populated-ness proxies for record completeness, and completeness in turn correlates with being a recognised returning customer. This is a specific, measured instance of a general hazard in self-selecting feature spaces: when a platform deactivates sparse predictors and groups correlated ones, the missingness structure those predictors carried is not removed but re-emerges, proxied by whichever surviving feature correlates with data availability. The finding does not undermine the explanation methods, all three of which faithfully report that the model relies on the feature; it shows that faithful attribution and behavioural meaning are distinct, and that the gap between them is exactly where a data artefact can hide. Taken together, the two results illustrate the wider payoff of the validation chapters. Once the explanations are trusted to represent the model, they stop being a reporting tool and become a diagnostic: in a richer model they would point marketing toward the signals worth acting on, and here they point the CDH owners toward what the model is not yet capturing, namely thin personalisation for this offer and a predictor that tracks data completeness rather than behaviour. Either way the value comes from the same place, that the patterns can be read as real properties of the model rather than as noise introduced by the explanation method.

4.3 Academic Implications

The thesis contributes at two related levels to the XAI evaluation literature.

The first contribution is methodological: a structured instability-attribution protocol that decomposes observed explanation ranking shifts into three sources, using a pair of effect-size diagnostics applied directly to the propensity score distribution and the active-snapshot set. The decomposition is approximate rather than causal, and the chapter is explicit on that point, but it makes a set of distinctions that are usually left merged together in XAI evaluation studies. The combination of the Kolmogorov–Smirnov statistic on the model output distribution alongside Jaccard overlap on the set of active model-version identifiers has not, to the author's knowledge, previously been operationalised inside a CDH evaluation framework.

The second contribution is empirical: a systematic comparison of SHAP, LIME, and a decision-tree surrogate applied to Pega ADM propensity scores under production conditions, including continuous online updating, observational exposure-biased data, categorical-heavy feature spaces, and restricted model access. Standard XAI benchmarks evaluate methods on i.i.d. static datasets with full model access (Molnar, 2025); this thesis reports XAI behaviour under the inverse of those assumptions. The contribution is to document what changes, what does not, and which evaluation choices matter most when those assumptions are dropped.

A further methodological observation concerns the choice of surrogate architecture. The candidate-model comparison shows that an architecture-matched Naive Bayes surrogate is not the best approximator of an online-updating Naive Bayes black-box model. Matching a surrogate to the black box's own model class is an intuitive default, but the comparison here shows it is not the best choice in this setting; whether this holds more generally is left open and would benefit from a dedicated study against the surrogate-modelling literature.

4.4 Practical Implications

For organisations deploying explainability tooling on continuously-updating decisioning models like the one studied here, five implications follow.

First, the choice of explanation method matters operationally, not only academically. Decision-tree surrogates are often preferred in governance contexts because they are interpretable end-to-end. The results in this thesis show that the same surrogate can produce substantially different feature rankings on subgroups of the same population even when the underlying model has not changed. Practitioners who rely on per-subgroup decision-tree explanations for compliance or audit purposes should be aware that the resulting story can be inconsistent across populations for reasons that have nothing to do with model behaviour. SHAP and LIME, despite their own theoretical limitations, are consistently more stable in the present setting.

Second, the instability-attribution diagnostic is directly deployable for monitoring. The two effect-size statistics underlying the diagnostic, the Kolmogorov–Smirnov statistic on logged propensity and the Jaccard overlap on the active model-version set, are cheap to compute from existing decisioning logs and require no access to ADM internals. A monitoring pipeline that tracks these alongside per-window explanation rankings would flag instability events and indicate, before any deep diagnostic work, whether the likely source is data drift, model turnover, the explanation method itself, or some combination.

Third, the surrogate-based explanation pipeline as a whole is a workable approach for practitioners without access to ADM internals. The Pega CDH platform does not currently expose a live scoring API for offline analysis. The pipeline operationalised in this thesis, surrogate fitted on logged $(\mathbf{X}, p)$ tuples, fidelity assessed against the same logs, attributions explained on the surrogate, demonstrates that meaningful explanation work is possible inside that constraint. It does not replace direct ADM access where that is available; it provides a defensible second-best for the common case where it is not.

Fourth, the feature-driver findings reframe where personalisation effort is best spent for this offer. Because the propensity score is dominated by contact recency and frequency rather than by customer attributes (Section 3.5), the score responds more to the cadence of the outbound programme than to customer segmentation. A practical consequence is that a contact-frequency or fatigue-management policy, governing how recently and how often a customer is emailed, is likely to shift propensities more reliably than investment in richer customer features. This holds, however, only because the customer-level predictors are currently too sparse for Pega to activate them, so the score falls back on recency and fatigue largely by default. Improving the customer-level data pipeline, so that these predictors carry enough signal to be retained, is therefore a worthwhile investment in its own right: the present reliance on contact cadence is better read as a sign of that data gap than as evidence that customer attributes do not matter. If marketing stakeholders ask which customer features the model relies on for this offer, the answer is simply: none. No customer attribute was retained as a predictor.

Fifth, high-importance predictors should be checked against their own populated-ness before their attribution results are relied on. The staff-standby case shows that in a self-selecting feature space a predictor can rank highly because its presence proxies for record completeness rather than because it carries behavioural signal. This is cheap to detect, by comparing the missingness pattern of a top predictor against the availability of the deactivated features. Doing so guards against a specific failure: signing off a model on the basis of an explanation that is faithful to the model yet misleading about the real world, because the feature it highlights is really standing in for which records happen to have data. More broadly, it argues for recording which predictors were deactivated and which were grouped together as correlated, and treating that information as a standard part of any CDH explanation review, since it determines which signals a surviving feature may silently be standing in for.

4.5 Limitations

A number of limitations qualify the interpretation of these findings. The most fundamental assumption underlying the whole design is that the CatBoost surrogate is a faithful enough stand-in for the production model to be worth explaining; this is tested rather than assumed, through the fidelity assessment in Chapter 3, and the limitations below should be read against that assumption.

The most fundamental is the double approximation that the explanation pipeline introduces. SHAP and LIME do not explain the production model f directly; they explain the surrogate g that approximates f. Explanation validity is therefore bounded by surrogate fidelity, and any region of the feature space where g diverges from f is a region where the resulting attributions may not faithfully describe Pega's behaviour. The fidelity assessment in Chapter 3 establishes that g approximates f well in aggregate for three of the four variants studied, but does not certify pointwise fidelity for every input. The Cartrawler variant in particular sits below the level at which downstream attribution claims should be made with confidence; its results are reported and replicated, but should be given less weight than the others. The decision-tree surrogate sidesteps one layer of approximation by fitting directly on p, but at the cost of a much coarser representation of f and the structural ranking instability documented above. Neither path is approximation-free, and the trade-off between fidelity layers and explanation interpretability is itself part of the methodological lesson of the thesis.

A related limitation is the encoding mismatch between the surrogates. The decision tree is implemented through scikit-learn and therefore uses ordinal encoding of categorical features, whereas CatBoost handles them natively. The decision-tree-versus-SHAP gap reported as δ_e^{DT} partly reflects this encoding difference, and the methodology chapter is explicit that the gap is best interpreted as a composite of explainer sensitivity and encoding sensitivity rather than as a pure explainer effect.

A second class of limitations concerns the data and setting. The study is conducted within a single European airline, on the Email and Outbound channel only, and on a small portfolio of offers concentrated around luggage upsells with two third-party partner offers as replication. The replication design provides a meaningful generalisation check across offers within one CDH installation but does not constitute cross-industry or cross-channel validation. Findings should be interpreted as evidence about XAI method behaviour under the specific conditions of this production environment, with implications for practitioners operating in similar settings rather than as universal claims about explanation methods.

A third class of limitations concerns observational structure. All evaluation data is drawn from logged decisioning events, with no ability to query the live ADM scorer on counterfactual or held-out inputs. Exposure bias is intrinsic to the data: customers respond only to offers they were shown, and both the propensities and the explanations of those propensities are conditioned on the historical decisioning policy (Lundberg, 2018). Running the surrogate-and-attribution pipeline live, alongside the production ADM and on the same inputs as they are scored in real time rather than only on historical logs, would tighten some of the current claims, but this was outside the scope of the project given the access constraints.

A further limitation concerns the instability-attribution diagnostic itself. The decomposition into distribution shift, model-version churn, and explainer sensitivity is a structured approximation, not a causal identification: the three sources are estimated from effect-size statistics rather than isolated by intervention, and δ_d and δ_m are correlated, since the incoming response data that shifts the feature distribution also drives the online model updates. The diagnostic should be read as an indicative decomposition that separates sources usually merged into a single instability figure, not as a precise statement of cause.

Two narrower limitations are worth flagging. The Jaccard overlap on model-version identifiers proxies snapshot turnover rather than parameter shift directly: Pega ADM issues a new identifier on every micro-update, so two functionally identical snapshots can carry different IDs. This is mitigated by the structural property that same-window splits draw from the same active snapshot pool, but it remains an imperfect proxy. Separately, LIME importance rankings are aggregated across instances by averaging per-instance linear coefficients, which assumes coefficient comparability across the feature space; the conclusions on LIME stability should be read with that assumption in mind.

4.6 Future Research

Several extensions follow naturally from the results and the limitations.

A direct next step is to incorporate Pega ADM's own predictor importance signal as an external reference. The platform exposes per-predictor univariate AUC and bin-level lifts through its

Health Check predictor overview. These quantities are not, strictly, explanations of an individual scoring event, but they do provide a model-internal view of which predictors carry evidential weight, and they can serve as a soft reference ranking against which the surrogate-based SHAP, LIME, and decision-tree attributions can be compared. Such a comparison was deliberately held outside the scope of this thesis to preserve the black-box framing on which the methodology rests, but it would constitute a valuable validation step in subsequent work, particularly for the cross-offer replication where surrogate fidelity varies.

A second direction is cross-channel and cross-industry replication. The present results cover Email and Outbound only and one airline; replicating the protocol on web, push, or mobile channels, and on CDH installations in other industries such as telecommunications, retail, or financial services, would test whether the patterns documented here are properties of CDH systems generally or of the airline luggage portfolio specifically.

A third direction concerns longitudinal evaluation. The temporal splits in this study compare early and late halves of a single observation window. A longer-horizon study, with multiple sequential splits across months or quarters, would show whether explanation rankings drift gradually or jump around erratically over time, and whether the instability-attribution diagnostic stays consistent across rolling windows. It would also make it possible to study seasonal effects, which are pronounced in airline demand, on both the propensity distribution and the resulting explanations, and to add formal concept-drift tests on the feature-level distributions alongside the propensity-output diagnostic used here.

Counterfactual explanation methods and real-time explanation systems are natural complementary directions. Counterfactual approaches (Molnar, 2025) provide explanations of the form "the minimal change to the input that would have changed the decision"; they are a complement to the feature-attribution methods evaluated here, with different fidelity and stability properties that this study does not address. A real-time explanation system, by which I mean one that produces and logs explanations alongside live scoring events rather than retrospectively, would address the observational-data limitation directly.

Finally, two model-side questions follow directly from the findings. The first is whether a richer model class would actually predict better in production. Pega can run an alternative model, Adaptive Gradient Boosting, in shadow mode, that is, scoring alongside the live model without driving any decisions, and comparing it against the online Naive Bayes would test this directly. The candidate-surrogate comparison here shows only that a flexible gradient-boosted *surrogate* approximates Pega's logged propensities better than an architecture-matched Naive Bayes surrogate can; whether that gap would translate into real predictive headroom under live deployment cannot be settled from the present data. The second question follows from RQ5. Because the model currently scores recency and contact fatigue rather than customer identity, a natural next step is to repair the customer-level feature pipeline, check whether the now-populated customer attributes are retained as predictors, and test whether they predict click-through better than the recency and frequency signals that dominate today. A related experiment would be to add an explicit missingness indicator and see whether it absorbs the importance currently attached to the staff-standby proxy.

Chapter 5

Conclusion

This thesis evaluated post-hoc explanation methods applied to a continuously-updating offer-level propensity model in a production airline Customer Decision Hub, and proposed a structured diagnostic for attributing observed explanation instability to its dominant source. The five research questions are answered directly below.

5.1 Surrogate Fidelity (RQ1)

A CatBoost surrogate trained on logged Pega ADM propensity scores reproduces those scores with high fidelity on three of the four offer variants studied. Rank-based agreement is the operationally relevant property here, because Pega CDH consumes propensities through prioritisation rather than as calibrated probabilities. It is strong on the primary L5B15 luggage variant and on the CLUG and BookingDotCom replications, and weaker on Cartrawler. The surrogate provides a sufficiently faithful approximation of the black-box model to serve as a valid explanation target on the three variants where rank fidelity is high; Cartrawler-specific results are reported under the explicit fidelity caveat.

5.2 Explanation Stability (RQ2)

Across temporal, route, and customer-culture splits on L5B15, SHAP produces the most stable feature rankings, with full-ranking Spearman correlations close to unity in every comparison. LIME tracks SHAP closely, with one exception on the route pair with neither shared origin nor destination, where local perturbation sampling introduces additional variance. A shallow decision-tree surrogate fitted directly on the propensity scores is substantially less stable than both, particularly on subgroup splits, and the bootstrap validation confirms that the between-split disagreement exceeds within-split fitting noise on three of the four subgroup comparisons.

5.3 Instability Attribution (RQ3)

When explanation rankings shift between sub-populations, the dominant source depends on the type of split. On temporal splits both the data, in the form of the propensity distribution, and the model, in the form of the active model-snapshot set, shift substantially, and the diagnostic classifies these as jointly driven by data drift and model churn. On same-window subgroup splits, where both external sources are constrained to be small by design, the dominant remaining source is the explanation method itself. The implication is that under the operationally realistic condition in which data and model are approximately stable, the choice of explanation method materially shapes which features appear to drive scoring.

5.4 Cross-Offer Replication (RQ4)

The patterns documented for L5B15 replicate across three further offer variants on temporal and route splits, with classifications changing only on borderline values of the distribution-shift diagnostic and with the decision tree's refit gap remaining the largest numerical contributor throughout. The culture split is the one place where L5B15 does not replicate: the three replication variants exhibit substantial culture-conditional propensity shifts that the luggage upsell does not. The evaluation method therefore carries over from one offer to another. Which particular feature ends up driving the instability, by contrast, depends on the specific offer and its customers.

5.5 Feature Drivers (RQ5)

For the L5B15 offer, the propensity score is driven by contact recency and frequency and by the context of the booked trip, while customer-level attributes play no role because none are retained as active predictors. All three explanation methods agree on this at the feature-group level, so the finding is robust to the explainer-stability concerns of RQ2 and RQ3. The model is, in this sense, scoring how recently and how often the customer was contacted more than who the customer is. This points marketing effort toward contact-cadence management rather than richer segmentation, with the caveat that the importance figures show what the model reacts to, not a guaranteed lever for changing outcomes. The other side of this is equally valid: the model ignores customer attributes only because none were retained as predictors, so a reasonable response is to invest in the customer-level data, by repairing its collection or adding new customer features, and then test whether they become predictive of click-through. The finding says where the model looks today, not where it could usefully look once the data supports it. A separate, cautionary finding concerns one of the model's most influential predictors, the staff-standby flag. It is important not because staff-standby status carries any behavioural signal, but because whether the field is populated proxies for how complete a customer's record is. This is a reminder that in a self-selecting feature space, a faithful attribution and a behaviourally meaningful one are not the same thing. High-importance predictors should therefore be checked against their own missingness before they are trusted in governance.

5.6 Final Synthesis

The principal contribution of this thesis is a structured evaluation protocol for post-hoc explanation methods applied to continuously-updating propensity models under the conditions that characterise production CDH environments: observational decisioning logs, restricted black-box access, categorical-heavy feature spaces, and continuous online learning. The protocol combines surrogate-based fidelity assessment with a three-way instability-attribution diagnostic that decomposes observed ranking shifts into distribution shift, model-version churn, and explainer sensitivity. Applied to a portfolio of four offer variants from a single CDH installation, the protocol reveals a consistent pattern: SHAP and LIME deliver stable, fidelity-grounded attributions; the decision-tree surrogate is the least stable explanation method across populations even when the underlying model has not changed; and explainer sensitivity is the dominant source of instability under controlled same-window conditions. For practitioners operating in CDH environments without direct model access, the protocol provides a defensible second-best path to model-explanation evaluation. For the wider XAI evaluation literature, it offers empirical evidence on how feature-attribution methods behave when the standard i.i.d., full-access, static-model assumptions are dropped. Alongside the protocol, the case study shows what that validation is ultimately for. Once SHAP and LIME are trusted to represent the model, their attributions become a lens on what the production model has actually learned. In a richer model that lens would motivate the treatments and offers marketing builds on top of the scores. Here it mostly exposes modelling gaps, which is no less useful to the stakeholders who own the model. Two such findings stand out: that the offer-propensity model scores contact recency and frequency rather than customer identity, which says personalisation for this offer is thinner than its owners likely assume; and that a self-selecting feature space can elevate a predictor whose importance encodes data missingness rather than behaviour, a governance hazard that is cheap to detect once it is known to look for. Both are credible only because the validation chapters established that they reflect the model rather than the method used to explain it.

Chapter 6

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Chapter 7

Appendix: Additional Detail

A.1 Feature Group Missingness Profile

Table A.1 reports the number of active predictors and mean missingness per feature group for each analysed offer variant. Across variants, contact history is the only group with substantial missing data: the mean fraction of missing values ranges from 41.6% on L5B15 to 57.4% on BookingDotCom, reflecting customers with limited prior contact history on the email and push channels. Booking context and flight data are nearly complete (under 2.5% missing across all variants), and the customer-attribute features that appear only in the BookingDotCom variant are fully populated.

The number of active predictors per group varies across variants because Pega ADM selects predictors independently per model based on each predictor's evidential contribution; BookingDotCom is the only variant whose final model includes customer-attribute predictors. Scoring parameters appear with a dash in the missingness column because they are populated at scoring time rather than read from a customer record, so the underlying parquet column for the CatBoost categorical encoding has no row-level absence to count.

Table A.1: Feature groups, descriptions, and mean missingness profile across all analysed offer variants. Active predictor counts (n) vary because Pega ADM selects predictors independently per model based on evidential contribution. Customer attributes appear only in the BookingDotCom variant. BDC = BookingDotCom; CTL = Cartrawler.

		L5B15		CLUG		BDC		CTL	
Feature group	Description	n	Miss.	n	Miss.	n	Miss.	n	Miss.
Booking context	Fare class, journey type, product class, culture	9	0.5%	7	0.7%	7	0.6%	8	1.0%
Flight data	Route, operator, aircraft type	2	1.1%	4	1.5%	4	2.0%	4	2.1%
Contact history	Contact frequency and recency across channels	11	41.6%	10	44.6%	18	57.4%	10	54.3%
Scoring parameters	Parameters passed at scoring time	1	–	1	–	1	–	3	–
Customer attributes	Demographics and loyalty profile	–	–	–	–	3	0.0%	–	–

A.2 Cross-Validation Depth Selection: L5B15

Table A.2 reports the five-fold cross-validation Spearman ρ_S over depths 1 to 15 for the L5B15 CatBoost surrogate. The CV-maximum is reached at depth 12. Applying Breiman's 1-SD rule, depth 10 (bold) is selected as the smallest depth whose mean $\bar{\rho}_S$ falls within one standard deviation of that maximum. The plateau between depths 10 and 13, followed by a marginal decline at depths 14 and 15, confirms that the grid upper bound of 15 is sufficiently wide to capture the true optimum rather than truncating the curve at its edge.

Table A.2: Five-fold cross-validation Spearman ρ_S over depths 1–15 for the L5B15 CatBoost surrogate. CV-maximum is at depth 12; the 1-SD parsimony rule selects depth 10 (bold).

Depth	$\bar{\rho}_S$	σ_{ρ_S}
1	0.830	0.006
2	0.892	0.001
3	0.903	0.002
4	0.907	0.002
5	0.911	0.003
6	0.913	0.004
7	0.917	0.003
8	0.920	0.002
9	0.922	0.003
10	**0.924**	**0.003**
11	0.924	0.002
12	0.925	0.002
13	0.925	0.003
14	0.924	0.002
15	0.924	0.002

A.3 Model Version Churn: Cross-Check on the AUC Distribution

The δ_m proxy used in the main analysis is the Jaccard overlap on the set of active model-version identifiers (Section 2.5), which answers the snapshot-overlap question directly. As an independent cross-check, three further candidate metrics were evaluated on the model's logged AUC (`modelPerformance`) distribution. PSI on the AUC distribution produced values between 9 and 15 on temporal splits, well above the standard significance threshold of 0.25, making the absolute values uninterpretable and the standard thresholds inapplicable to this discretised distribution. Mean AUC difference was near zero across all splits, including temporal splits where substantial model churn is expected, and therefore failed to distinguish controlled from uncontrolled conditions. Of the three, only the Kolmogorov–Smirnov statistic on the AUC distribution cleanly separates temporal splits, where model churn is expected, from route and culture splits, where it is not. Table A.3 reports all three metrics across all splits. Their verdict agrees with the Jaccard-based classification: temporal splits show clear churn while same-window route splits do not. The AUC-distribution metrics therefore corroborate the Jaccard proxy rather than replace it.

Table A.3: Comparison of candidate metrics for model version churn (δ_m) across all splits. AUC PSI uses equal-frequency bins ($B = 5$) on the modelPerformance distribution. $\Delta\mu_{\text{AUC}}$ is the absolute difference in mean AUC between splits. KS is the two-sample Kolmogorov–Smirnov statistic on the AUC distribution. KS cleanly separates temporal splits (KS $\approx$ 0.33–0.50, $p \approx 0$) from route splits (KS $\approx$ 0.04, $p > 0.39$), making it the most informative of the three AUC-based candidates. These metrics corroborate the Jaccard overlap on model-version identifiers, which is used as the δ_m proxy in the main analysis.

Split	AUC PSI	$\Delta\mu_{\text{AUC}}$	KS	KS p
Temporal splits				
L5B15 temporal	14.076	0.001	0.433	<0.001
CLUG temporal	9.076	0.001	0.335	<0.001
BookingDotCom temporal	14.966	0.006	0.499	<0.001
Cartrawler temporal	14.118	0.004	0.490	<0.001
Route splits (L5B15)				
ORY→NCE vs ORY→OPO	0.013	0.000	0.040	0.394
ORY→OPO vs NCE→ORY	0.007	0.000	0.037	0.511
NCE→ORY vs TLS→ORY	0.007	0.000	0.035	0.625

A.4 Surrogate Model Selection

This appendix reports the supplementary surrogate-model evaluation used during model selection. No single candidate dominated all fidelity metrics: Random Forest achieved the highest Spearman rank correlation, LightGBM achieved the lowest KS statistic, and CatBoost achieved the strongest pointwise fidelity and the highest Kendall τ. CatBoost was selected as the final surrogate because it provided the strongest overall balance across the metric suite while additionally supporting exact TreeSHAP computation and native handling of categorical predictors.

This distinction is methodologically important for the interpretation of downstream explanations. Applying SHAP to a non-tree surrogate would require KernelSHAP, a model-agnostic approximation based on Monte Carlo sampling. In the present setting, explanations already inherit approximation error from the surrogate itself, since the true Pega ADM model is inaccessible. Introducing an additional sampling-based approximation layer at the explanation stage would compound these sources of inexactness, making it more difficult to isolate whether observed instability originates from the surrogate or from the explainer. Using CatBoost instead allows SHAP values to be computed exactly through TreeSHAP, eliminating this second approximation layer. CatBoost was additionally preferred over alternatives requiring ordinal encoding because its native categorical handling preserves the original categorical representation used during model fitting, avoiding arbitrary numeric orderings that could otherwise distort SHAP attributions.

Table A.4: Per-variant CatBoost surrogate fidelity on the held-out 20%. CV-max is the depth maximising 5-fold CV Spearman ρ; Depth is the 1-SD parsimony pick actually used for the final model fit.

Variant	n	CV-max	Depth	R^2	RMSE	Spearman ρ	Kendall τ	KS statistic
L5B15	31,709	12	10	0.927	0.0026	0.933	0.798	0.154
CLUG	16,984	10	7	0.954	0.0026	0.919	0.779	0.182
BookingDotCom	19,866	11	9	0.792	0.0002	0.949	0.811	0.072
Cartrawler	20,645	13	12	0.911	0.0005	0.615	0.457	0.313

Table A.5: Predictive fidelity to Pega ADM propensity scores per variant, on the held-out 20%. The Decision Tree is fit directly to the logged scores; the CatBoost row applies to both SHAP and LIME, which explain the same surrogate and therefore share its predictive fidelity to Pega. R^2 and RMSE measure absolute-value agreement; Spearman ρ and Kendall τ measure rank agreement; the Kolmogorov–Smirnov statistic measures distributional agreement (lower is better, 0 = identical distributions).

Variant	Method	R^2	RMSE	Spearman ρ	Kendall τ	KS
L5B15	Decision Tree	0.604	0.0061	0.924	0.778	0.230
L5B15	CatBoost (SHAP + LIME)	0.927	0.0026	0.933	0.798	0.154
CLUG	Decision Tree	0.871	0.0044	0.906	0.765	0.282
CLUG	CatBoost (SHAP + LIME)	0.954	0.0026	0.919	0.779	0.182
BookingDotCom	Decision Tree	0.226	0.0003	0.550	0.400	0.254
BookingDotCom	CatBoost (SHAP + LIME)	0.792	0.0002	0.949	0.811	0.072
Cartrawler	Decision Tree	0.170	0.0015	0.318	0.226	0.463
Cartrawler	CatBoost (SHAP + LIME)	0.911	0.0005	0.615	0.457	0.313

Table A.4 reports the per-variant CatBoost fidelity and selected model depths actually used in the final fits, complementing the architecture comparison in Table 5 of Section 3.1.

Table A.5 reports fidelity at the explainer level rather than the surrogate level: the Decision Tree is fitted directly on Pega's logged scores, whereas SHAP and LIME explain the CatBoost surrogate and therefore share its predictive fidelity to Pega. The Decision Tree's RMSE is materially larger than CatBoost's on every variant, reflecting the cost of approximating a continuous-valued target with a small set of axis-aligned splits.

A.5 PSI Bin-Sensitivity Analysis

The choice of the Kolmogorov–Smirnov statistic over the Population Stability Index for δ_d rests on PSI's bin-sensitivity on Pega ADM's quantised propensity output. Pega applies an isotonic calibration step that maps log-odds scores to propensities via a step function: instances in the same score bin all receive the same propensity value, so the empirical distribution clusters on a small set of recurring values rather than spreading continuously across $[0, 1]$. Increasing the PSI bin count then forces these clustered values to split unevenly between the reference and comparison halves, inflating PSI artificially even when the underlying distribution has not

Table A.6: Propensity-PSI sensitivity to bin count. Each row corresponds to a different choice of PSI bin count (equal-frequency on the reference half); columns are the splits used in the main analysis. PSI values rise systematically with bin count, particularly for the temporal split, because Pega ADM's isotonic calibration produces a step-function propensity: increasing the bin count splits clustered values unevenly between the two halves and inflates PSI artificially. This motivates the choice of the bin-free KS statistic on propensity as the δ_d metric in the main analysis. Route pair 1: ORY→NCE vs ORY→OPO; Route pair 2: ORY→OPO vs NCE→ORY; Route pair 3: NCE→ORY vs TLS→ORY; Culture: fr-FR vs nl-NL.

bins	Temporal	Route pair 1	Route pair 2	Route pair 3	Culture
3	1.259400	0.032000	0.034500	0.031900	0.037000
4	1.009500	0.050400	0.018900	0.027000	0.044800
5	4.957900	0.040100	0.054000	0.025100	0.068300
6	4.119800	0.041400	0.061300	0.032200	0.062200
7	3.649000	0.054100	0.064300	0.055300	0.098000
10	6.611200	0.049700	0.075600	0.040900	0.111000

changed.

Table A.6 reports propensity-PSI computed at six bin counts ($B \in \{3, 4, 5, 6, 7, 10\}$) for every split used in the main analysis. PSI values rise systematically with bin count, particularly for the temporal split, confirming the bin-sensitivity concern. The ranking of splits by PSI is preserved across bin choices, so PSI remains informative for relative comparison, but the absolute value cannot be threshold-tested reliably. The KS statistic adopted in the main analysis avoids this entirely: it operates on the empirical CDFs without any binning or smoothing parameter and returns the same effect-sizes.

A.6 Jaccard Threshold Sensitivity

The model-churn threshold $J_v \leq 0.10$ used in the main analysis is defended here by inspecting where the observed J_v values actually sit. Figure A.1 plots the five L5B15 split values on the $[0, 1]$ line alongside four candidate thresholds. The values are strongly bimodal: the temporal split sits at $J_v \approx 0$, while route and culture splits cluster near $J_v \approx 0.5$. This bimodality is itself a methodological observation: same-window splits land near 0.5 by construction because Pega's online Naive Bayes issues a fresh model version on every micro-update, so two same-week sub-populations naturally see only about half their snapshot IDs in common even when the underlying model parameters have barely moved. The empty interval between the two clusters means any threshold between roughly 0.05 and 0.40 would assign the same Primary-source label to every split. The chosen value $\tau = 0.10$ sits near the lower edge of that band, giving maximum margin from the in-window cluster while still classifying every temporal split as significant churn.

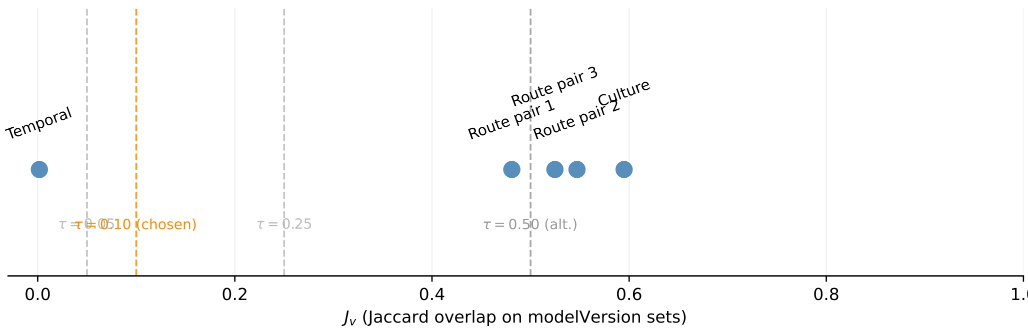

Figure A.1: Observed J_v values per L5B15 split, with candidate thresholds τ marked. The strongly bimodal distribution (temporal near 0, route and culture splits near 0.5) places the chosen $\tau = 0.10$ in an empty band where classifications are invariant.

Table A.7: Within-sample DT reproducibility versus observed between-split ρ_S^{DT}. For each sub-sample the DT surrogate was re-fitted on identical data with $N = 10$ random seeds, yielding 45 pairwise Spearman ρ values per sub-sample. Within (mean) is the average of those means across the two sub-samples of each split. Gap = Within − Between. Classification: > 0.10 genuine signal; 0.03 to 0.10 small effect; ≤ 0.03 indistinguishable from refit noise.

Split	Within (mean)	Between	Gap	Classification
ORY→NCE vs ORY→OPO	0.824	0.510	+0.314	Genuine signal
ORY→OPO vs NCE→ORY	0.789	0.584	+0.206	Genuine signal
NCE→ORY vs TLS→ORY	0.802	0.817	-0.015	Within noise floor
Temporal (early vs late)	0.935	0.855	+0.080	Small effect
Culture (fr-FR vs nl-NL)	0.977	0.666	+0.310	Genuine signal
Routes (aggregate)	0.813	0.637	+0.176	Genuine signal

A.7 Decision Tree Bootstrap Reproducibility

Table A.7 reports within-sample DT reproducibility alongside the observed between-split Spearman rank correlation, for each L5B15 split. For each sub-sample the decision tree was re-fitted ten times with different random seeds, yielding 45 pairwise ρ values per sub-sample; the within mean reported is the average across the two sub-samples involved in each split. The gap between within and between ρ_{DT} separates instability attributable to genuine subgroup composition from random fitting noise. Three of the five splits show gaps above 0.10, classified as genuine signal. The NCE→ORY vs TLS→ORY split shows a negative gap, indicating that the between-split drop falls within the noise floor for that particular route pair.

A.8 Per-Split Top-10 Feature Rankings

Table A.8 lists the ten highest-ranked features per L5B15 split per explanation method, taken directly from the saved per-split rankings used in the stability analysis. The qualitative agreement between SHAP and LIME on the top features, and the broader disagreement of the Decision Tree with both, mirrors the quantitative pattern reported in the stability matrix (Table 6).

Table A.8: Top-10 feature attributions per L5B15 split per explanation method.

split	DT	SHAP	LIME
early	1. Email.Out.Pending.HistCount 2. Param.BundleName 3. Email.Out.Delivered.LastOut.Days 4. Email.In.Clicked.LastOut.Days 5. Push.Out.Pending.LastOut.Days 6. FlightInboundArrival 7. Email.Out.Pending.LastOut.Days 8. SeatNumber 9. Email.In.Pending.LastOut.Days 10. Journey	1. Param.BundleName 2. Email.Out.Pending.LastOut.Days 3. Push.Out.Pending.LastGroup 4. Email.Out.Delivered.LastOut.Days 5. SeatNumber 6. IsStaffStandBy 7. BookingMonth 8. Email.Out.Pending.LastGroup 9. Email.Out.Pending.HistCount 10. FlightInboundArrival	1. Param.BundleName 2. SeatNumber 3. IsStaffStandBy 4. BookingMonth 5. Email.Out.Pending.LastOut.Days 6. Email.Out.Pending.HistCount 7. Email.In.Clicked.LastOut.Days 8. Email.Out.Delivered.LastOut.Days 9. DestinationAirport 10. Language
late	1. Email.Out.Pending.HistCount 2. Email.In.Clicked.LastOut.Days 3. Param.BundleName 4. Push.Out.Pending.HistCount 5. Email.Out.Delivered.LastOut.Days 6. Email.In.Pending.LastOut.Days 7. SeatNumber 8. Email.Out.Pending.LastOut.Days 9. BookingMonth 10. FlightInboundArrival	1. Param.BundleName 2. Email.Out.Pending.LastOut.Days 3. Push.Out.Pending.LastGroup 4. Email.Out.Delivered.LastOut.Days 5. SeatNumber 6. Email.Out.Pending.LastGroup 7. BookingMonth 8. IsStaffStandBy 9. Email.Out.Pending.HistCount 10. FlightInboundArrival	1. Param.BundleName 2. SeatNumber 3. IsStaffStandBy 4. BookingMonth 5. Email.Out.Pending.LastOut.Days 6. Email.Out.Pending.HistCount 7. DestinationAirport 8. Email.In.Clicked.LastOut.Days 9. Email.Out.Delivered.LastOut.Days 10. FlightInboundArrival
ORY→NCE	1. Email.Out.Delivered.LastOut.Days 2. Param.BundleName 3. Push.Out.Pending.HistCount 4. Email.Out.Pending.LastOut.Days 5. Email.In.Clicked.LastOut.Days 6. FlightInboundArrival 7. BookingMonth 8. IH.Event.Outbound.RealTimeEvent.HistCount 9. Journey 10. Push.Out.Pending.LastGroup	1. Param.BundleName 2. Email.Out.Pending.LastOut.Days 3. Push.Out.Pending.LastGroup 4. SeatNumber 5. Email.Out.Delivered.LastOut.Days 6. Email.Out.Pending.LastGroup 7. IsStaffStandBy 8. Email.Out.Pending.HistCount 9. BookingMonth 10. FlightInboundArrival	1. Param.BundleName 2. SeatNumber 3. IsStaffStandBy 4. BookingMonth 5. DestinationAirport 6. Email.Out.Pending.LastOut.Days 7. Email.Out.Pending.HistCount 8. Email.In.Clicked.LastOut.Days 9. Email.Out.Delivered.LastOut.Days 10. Email.Out.Pending.LastGroup
ORY→OPO	1. Email.Out.Pending.LastOut.Days 2. Param.BundleName 3. Email.In.Pending.LastOut.Days 4. Email.Out.Delivered.LastOut.Days	1. Param.BundleName 2. Email.Out.Pending.LastOut.Days 3. Push.Out.Pending.LastGroup 4. Email.Out.Delivered.LastOut.Days	1. Param.BundleName 2. SeatNumber 3. IsStaffStandBy 4. BookingMonth 5. Email.Out.Pending.LastOut.Days 6. DestinationAirport

Appendix A

Appendix on Generative Artificial Intelligence Use

Generative AI was used throughout this thesis in the following ways.

Research design and structuring. Claude (Anthropic) was used to discuss and refine the research design, sharpen research questions, and structure the methodology chapter. AI-generated suggestions were critically evaluated and revised before incorporation.

Writing and editing. Claude was used to draft, restructure, and improve prose across all chapters. All AI-generated text was reviewed, adjusted, and rewritten where necessary to reflect the author's own reasoning and conclusions.

Code assistance. Claude and GitHub Copilot were used to assist with writing Python notebook cells for data ingestion, EDA, modelling, and visualisation. All code was reviewed, tested, and adapted by the author.

LaTeX formatting. Claude was used to generate and debug LaTeX code for tables, figures, and tikz diagrams.

Representative prompts used include:

- "Critically evaluate my introduction draft and identify inconsistencies with the actual data constraints."
- "How should I structure the methodology section for a thesis comparing XAI methods on a production propensity model?"
- "What does a problem formulation section in formal notation for a black-box surrogate explanation pipeline look like"
- "Generate a TikZ figure showing the full methodology pipeline from data to instability attribution."

All final decisions regarding content, interpretation, and conclusions are the author's own. The use of generative AI did not affect the empirical analysis or the validity of the results.